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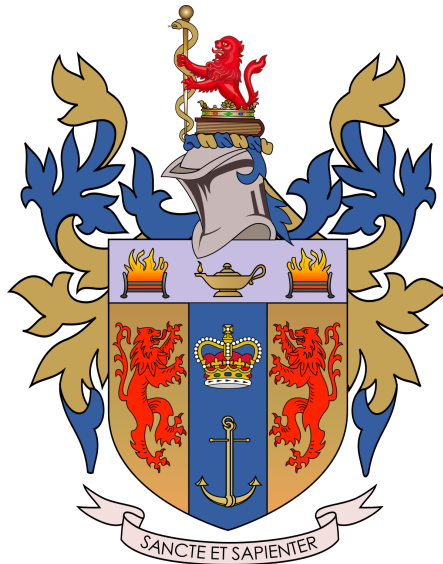
Regular and Logarithmic Conformal Invariance in Polymer Gelation

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Abstract

In this thesis, we attempt to find a closed set of fusion rules for critical polymer gelation for both the rational and logarithmic conformal field theories. We find the spectrum to be incomplete in the rational case, and infinite in the logarithmic case which is in good agreement with recent numerical results in the literature.

Dedicated to my brother Mikhail, without whom I would not be here today.

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1 Introduction

Conformal field theory (CFT) is one of the cornerstones of modern theoretical and mathematical physics, and has shown considerable success in high energy theory, and statistical mechanics. The latter of these successes was spearheaded by Belavin, Polyakov and Zamolodchikov in their seminal paper [2], which developed a framework allowing for the exact solution of a wide classification of statistical models in $d = 2$. However, there are notable exceptions to the framework, most prolific of which are the $c = 0$ critical percolation model and the $c = -2$ critical random walk. Indeed, we shall concern ourselves with the former of the two, and attempt to resolve the associated issues by introducing logarithmic extensions of the theory.

The percolation model provides a good description of a wide range of self-arranging phenomena, one example of which is polymer gelation, whereby the formation of bonds between monomers leads to an abrupt change in viscosity of the macroscopic material. This process has important applications in industry, such as in oil drilling, where they are used to plug gaps in flow channels dug for extraction, with the purpose of improving oil recovery rates [19]. To model the gelation process, we will consider how bonds build up randomly on a square lattice with a variable probability, which at a critical value, form a giant connected component.

In the literature review conducted for this project, CFT was introduced and elucidated on to convey the fundamentals of the formalism in a primarily mathematical setting. Standard reference texts were used in order to stage a further investigation into minimal models, and some preliminary notions and prerequisites of the bootstrap were introduced. We will briefly re-introduce the main ideas of CFT with a focus on physical intuition, although an in-depth discussion is left to the appendix and the literature review so as to expedite a discussion of minimal models and logarithmic extension.

The primary goal of this work is to attempt to find a closed fusion table which fully describes critical percolation, and thereby the gelation process. In the first section of this thesis, we present some background on polymer gelation processes, and then go on to examine percolation in the context of the $Q \rightarrow 1$ 2D Potts model, allowing us to make a heuristic link between percolation and the $c = 0$ CFT. In the following section, the essential notions of rational CFT are introduced, alongside the framework of the minimal models. The short-comings of regular conformal field theory are then presented with regards to treatment of the percolation model $\mathcal{M}_{2,3}, c = 0$, which is done by attempting to derive a complete set of fusion rules for the theory. In the penultimate section, logarithmic

conformal field theory is introduced, and it's existence within the model is shown in order to naively attempt to solve problems associated with the rational CFT approach to the derivation of the fusion rules. Finally, we will conclude with some final remarks and present an outlook on future directions of research on critical percolation.

2 Polymers and Percolation

As stated in the introduction, this thesis will primarily concern itself with looking at Percolation in the context of gelation processes. We spend much of this chapter introducing the relevant language and mathematics of our present focus, before we show the correspondence between the limit of $Q \rightarrow 1$ of the Potts model, and the $c = 0$ 2D CFT.

2.1 Polymers, Gelation, and Graph Theory

We wish to couch our further discussion of percolation and graph theory within the language of polymers. To this end, we can define important properties and surrounding concepts which will make the physical import of our discussion more apparent, and provide intuition for the associated mathematics. A graph is defined formally to be a pair of sets, with one containing the constituent vertices, and the other containing edges between those vertices. Our introduction to polymers and gelation is taken primarily from [15, 27].

2.1.1 Polymers and Gelation basics

A polymer is defined to be a long flexible chain of repeat units called monomers, which are connected together by physical or chemical bonds. Examples of these are van der Waals bonds and covalent bonds respectively. Polymer Gels are networks of these flexible chains. Such structures are obtained through chemical or physical processes, and are dependent on the specific properties of the gel in question. As part of this process, polymer chains are usually suspended in a liquid, forming a solution. This is called the sol-phase of the solution. By then tuning the parameters to incite a chemical reaction, such as a condensation reaction, whereby Oxygen and Hydrogen atoms are removed from the original molecules to form water, allowing the remaining hydrocarbons to bond, these large chains can be built up and polymerized by covalent bonds until all the monomers are either directly or indirectly connected to one another, abruptly forming what is called a giant connected component, in the gel-phase. The formation of the giant connected component is associated with a sudden change in the viscosity of the associated material.

This abrupt change in the physical properties of the material is called a gelation transition, occurring at the gelation point, and the phase transition here is called a sol-gel transition.

We can consider now the existence of local versus global observables in polymers and specifically gelation. In polymers we consider that there is a major distinction between local properties which concern the characteristics of a singular monomer, and global properties which are defined over several monomers. Examples of the former are related to the properties of the monomer in question and are usually related to molecular properties such as size and reactivity. Examples of the latter include but are not limited to quantities such as the gyration radius R_G , which is the root mean square distance of polymer chain segments from the overall chain's center of mass. This property follows a scaling law [15]:

$$R_G = CN^\nu \quad (2.1)$$

Where C is some dimensionful proportionality constant, and N is the degree of polymerization, or the total number of monomers in the polymer chain. $\nu \simeq 3/5$ is a scaling exponent, and tells us how the observable scales at criticality. Another, more striking example of such an observable is the average degree of polymerization, weighted by molecular mass of each chain, N_w [15].

These non-local scaling properties give us cause to expect the appearance of non-local observables in the field theory of percolation, which is a strong indicator of non-unitarity in conformal field theory. Due to the geometric nature of percolation, an important subtlety to acknowledge is that boundary conditions, and therefore the finiteness of the system play integral roles in the associated dynamics. For example, the giant connected component become formally infinite in the bulk after the phase transition. The main focus of the rest of this thesis will be examining the behavior of critical percolation and it's associated CFT in the bulk, as boundary CFT is it's own rich field of study and research, which cannot be thoroughly explored due to time constraints, see [7, 9, 8] for further reference.

2.2 Percolation and the $Q \rightarrow 1$ Potts Model

We will consider a square lattice model of percolation here where no solvent is present, and each lattice site represents a monomer which can bond with it's nearest neighbors, represented by a bold line. An example of what a percolation system looks like below criticality is seen in Figure 1.

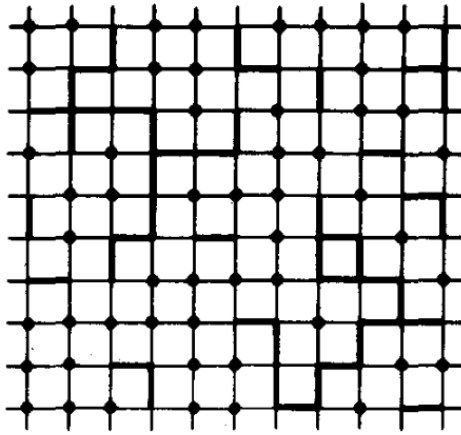


Figure 1: Lattice Model Example. Bold lines indicate a bond between sites. Image Credit: [15]

2.2.1 The Partition Function

Physics and geometry are often said to be quite inseparable fields, and it is partly for this reason that percolation is of interest to us, being that it is a uniquely geometric phase transition. Similar to the gelation process, percolation begins by assuming the presence of monomers at every site on a given $d = 2$ lattice, with the probability of there existing a bond between any two given monomers being globally p . We consider that varying our physical parameter is analogous to us varying the probability of there existing a bond between two adjacent monomer sites, and that there exists a critical probability p_c at which the gelation point is reached. We can make an expedient mapping of this problem to a more traditional spin-chain format by considering the $Q \rightarrow 1$ -state Potts model, where Q is an integer valued spin-state, following arguments by Cardy in this section [8]. The distinction here between $Q = 1$, and $Q \rightarrow 1$ is important, as the mapping we perform is from edges to spin-states. Regions of aligned spins here therefore correspond to connected components, and so the limit $Q \rightarrow 1$ is the so-called 'giant' connected component which is obtained after the phase transition. One of the unique traits of percolation is it's nature as a fundamentally geometric phase-transition, which leads to non-local observables and resulting non-unitarity as we shall see. The Potts model is a generalization of the Ising model, which extends the Ising-model framework by allowing more than 2 spin states. We now consider that for a given lattice site i , there are $\sigma_i = 1, \dots, Q$ possible spin-states where $Q \in \mathbb{Z}^+$. The Hamiltonian and partition functions for this model are given by:

$$H = -J \sum_{\langle i,j \rangle} \delta_{\sigma_i, \sigma_j}, \quad Z = \sum_{\Omega} e^{\beta J \sum_{\langle i,j \rangle} \delta_{\sigma_i, \sigma_j}}. \quad (2.2)$$

Where Ω is the set of all possible micro-state configurations, and the summation $\langle i, j \rangle$ represents a summation over nearest neighbors in the lattice. When summing over the entire configuration space Ω , there are two distinct subdomains, each corresponding to the bond not existing, which has probability $1-p$, and the bond existing which has probability p . This allows us to consider the re-arrangement:

$$Z = \sum_{\Omega} \prod_{\sigma_i, \sigma_j} ((1-p) + p\delta_{\sigma_i, \sigma_j}) \quad (2.3)$$

Where we set $p = 1 - e^{-\beta J}$ [8]. We consider that for a lattice of B bonds between monomers in the lattice, there must exist 2^B bond configurations, each configuration corresponding to whether the bond is open (the edge exists), or closed (the edge does not exist). Sites connected together form clusters. These clusters, are forced to have the same spin by the Kroenecker delta, and this allows us to further re-arrange our sum to a sum over the clusters in our system, which we will denote $\mathcal{C}$:

$$Z = \sum_{\mathcal{C}} p^{|\mathcal{C}|} (1-p)^{B-|\mathcal{C}|} Q^{N(\mathcal{C})} \quad (2.4)$$

Where the factor of Q indicates the overall spin state of the cluster, and $N(\mathcal{C})$ is the number of distinct clusters. We see now that the partition function is now a polynomial in Q [8]. The existence of a phase transition is shown in the appendix A. Formally in the above equation, we are summing over the configuration space of graphs, though we have not specified this notion. Graphs are characterized in full by matrices known as adjacency matrices, where the row and columns indicate graph vertices, or monomers in our case. The entries indicate the existence of a connection between them, which in this case are the bonds. As such when we are summing over the configurations, we are summing over all possible adjacency matrices which would characterize the polymer network in the bulk.

2.2.2 Criticality and $c = 0$ CFT

We have argued heuristically above that the phase transition occurs at the limit of $Q \rightarrow 1$ at the critical probability $p = p_c$, and the existence of a phase transition is shown in A. Therefore, we consider the asymptotic limit of criticality at $Q = 1$, which implies the existence of a single cluster $\mathcal{C} = 1$. In this way we see that our partition function approaches:

$$Z|_{Q=1} = \sum_{\mathcal{C}} p^{|\mathcal{C}|} (1-p)^{B-|\mathcal{C}|} = 1 \quad (2.5)$$

Exponent	Quantity	d=2
ν	Correlation Length: $\xi(p) \propto p - p_c ^{-\nu}$	4/3
γ	Cluster Density: $\chi(p) \propto p - p_c ^{-\gamma}$	43/18
σ	Cluster Size: $s_\xi(p) \propto p - p_c ^{-1/\sigma}$	36/91
β	Percolating Cluster: $P_\infty(p) \propto (p - p_c)^\beta$	5/36
τ	Cluster Number Density: $n(s, p) \propto s^{-\tau} \mathcal{G}(s/s_\xi)$	187/91

Table 1: Critical exponents of percolation in $d = 2$. Figure credit: [10].

Which is the sum over all cluster configurations weighted by the associated probabilities, and is therefore simply equal to 1. As such, our partition function is itself trivial, implying full conformal invariance of the theory. The theory of percolation is therefore associated with $c = 0$, as conformal symmetry can never be broken for a partition function which possesses a constant finite value. This association does make sense, as central charge is meant to arise from short-range conformal symmetry breaking arising from quantum effects, and is indicative of the rate of growth of the free energy of a system with size, as is seen in the Cardy formula[9]:

$$\Delta F = -\frac{1}{6}c\chi \ln(L) \quad (2.6)$$

Where L is some characteristic length of the system, and χ is the Euler characteristic of the manifold upon which the conformal field theory is defined. It is worth noting that $c = 0$ CFTs are not well-understood in general and are an active area of present research[20].

To examine conformal invariance of the system in greater detail, we must therefore consider the correlator rather than the partition function. However we return to such a calculation in later sections.

One of the successes of CFT is it's ability to reproduce exactly the scaling dimensions and so the critical exponents of order parameters in a statistical system such as the Ising model. An example is presented in the appendix C. In our case we are interested in percolation, so we wish to find the universal critical exponents thereof, which are helpfully found in the literature for $d = 2$ [10].

We focus the remainder of our efforts primarily on finding a well identified percolation observable which we hope to ground our search for a closed fusion algebra for percolation: the energy with conformal dimension $\Delta_\epsilon = 5/4$ [20]. There is a well documented relation between the scaling dimension and the corresponding critical exponent ν for the former[10]:

$$\Delta_\epsilon = \left(d - \frac{1}{\nu}\right) |_{d=2} = 5/4. \quad (2.7)$$

Where d is the number of spatial dimensions inhabited by the theory. We will see that we can clearly associate this observable with the conformal primary field $\phi_{2,1}$ with conformal weight $h, \bar{h} = \frac{5}{8} \implies \Delta_\epsilon = h + \bar{h} = \frac{5}{4}$. If we were examining a finite systems, we would also be interested in Cardy's crossing probability observable, telling us the probability of the giant connected component in a finite system having a contiguous connection between opposite boundary regions, which has been the corner stone of research surrounding percolation in finite systems[7].

We have laid out the physical and mathematical background for gelation and bond-percolation, as well as heuristically illustrated its association with a CFT with vanishing central charge. We also identified an observable with scaling dimension Δ_ϵ in $d = 2$, which is the energy observable of the system[20]. However, we will now see that this field lies outside the spectrum predicted by the framework of the rational minimal models, which will be the focus of the next section.

3 2d CFT

We will now re-introduce several essential components of 2D CFT originally presented in the literature review. The aim of this section is to show how CFT performs a root branch overhaul of QFT resulting from the imposition of extra symmetries. A physically motivated re-introduction of 2D CFT can be found in the appendix B.

3.1 Scale and Conformal Invariance

Dilational invariance is an example of a conformal symmetry, and it tells us that a prerequisite to conformal symmetry requires a theory to behave the same at all energy and length scales. It is this scale invariance which plays a critical role in CFT's ability to describe second order critical phenomena, where the correlation length of the theory blows up to infinity at criticality.

This scale invariance has several important corollaries, which we will now present. The first corollary is that conformally invariant theories must be both free and massless. To illustrate this, we consider the action of an interacting massive scalar field:

$$S = - \int d^d x (\partial_\mu \phi)(\partial^\mu \phi) + m^2 \phi^2 + g \phi^n. \quad (3.1)$$

Where g is a coupling constant. The action must be dimensionless for it to appear as an exponent in the path integral, so by inspection we can see that ϕ must have units of $(mass)^{\frac{d-2}{2}}$, where we call the exponent $\frac{d-2}{2} = \Delta$ the scaling dimension of the field.

Under a coordinate scaling $x \rightarrow x' = \lambda x$, we can see that in order to retain the action as dimensionless, the field ϕ must transform as[1]:

$$\phi \rightarrow \left(\frac{\partial x'}{\partial x} \right)^{-\frac{\Delta}{d}} \phi \quad (3.2)$$

A field transforming in this way is called a quasi-primary field [1]. We can now see how the various terms in the action transform under dilation to see how conformal invariance is broken by the mass and interaction terms:

$$S' = - \int d^d x' g'^{\mu\nu} \partial_\mu \phi' \partial_\nu \phi' + m^2 \phi'^2 + g \phi'^n \quad (3.3)$$

$$\begin{aligned} \text{New Kinetic term: } \int d^d x' g'^{\mu\nu} \partial_\mu \phi' \partial_\nu \phi' &= \int d^d x \lambda^d \Omega(x) \lambda^{-\frac{2\Delta}{d}} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi, \\ &= \int d^d x \Omega(x)^{-\frac{d}{2}+1+\Delta} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi, \\ &= \int d^d x \Omega(x)^0 g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi, \\ &= \int d^d x g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi = \text{Old Kinetic Term.} \end{aligned}$$

$$\text{New Massive Term: } \int d^d x' m^2 \phi'^2 = \int d^d x \lambda^d \lambda^{-\frac{2\Delta}{d}} m^2 \phi^2 \neq \text{Old Massive Term.}$$

$$\text{New Interaction Term: } \int d^d x' g \phi'^n = \int d^d x \lambda^d \lambda^{-\frac{n\Delta}{d}} g \phi^n \neq \text{Old Interaction Term.}$$

Where we have used the result that $\Omega(x)^{1/2} = \lambda$ for dilatation. This result is easily derived from the re-scaling of the metric.

The result $\Delta = \frac{d-2}{2}$ is a specific example here of an important fact of CFT, which is that fields, or order parameters, can be uniquely identified by their scaling dimension. This scaling dimension is related to observables, and critical exponents of a system, and so in $d = 2$ can be exactly calculated.

3.2 Correlation Functions

This scale invariance has important corollaries whereby all two and three point correlation functions of a theory are exactly determinable, evaluating to:

$$\langle \phi_1(x_1) \phi_2(x_2) \rangle = \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}}, \quad \begin{cases} C_{12} = 0, \\ \Delta_1 = \Delta_2. \end{cases} \quad (3.4)$$

and

$$\langle \phi_1(x_1) \phi_2(x_2) \phi_3(x_3) \rangle = \frac{C_{123}}{x_{12}^{\Delta_1 + \Delta_2 - \Delta_3} x_{23}^{\Delta_2 + \Delta_3 - \Delta_1} x_{13}^{\Delta_3 + \Delta_1 - \Delta_2}}. \quad (3.5)$$

Where $x_{ij} = |x_i - x_j|$. A proof of this can be found in B which closely follows arguments from [1]. This side-steps the often messy process of determining the correlation function of a theory via its equation of motion. A further examination of four and higher point functions will lead to the bootstrap methodology which reduces the computation of higher order functions to an algorithmic process, though we do not investigate that here.

As such, we have seen that in order to preserve conformal invariance of a theory, interactions and mass terms must be omitted from the lagrangian. Interactions are encoded in CFT by the operator product expansion.

3.3 The Operator Product Expansion

The operator product expansion (OPE) leverages the fact that 2 and 3-point functions are found automatically by conformal invariance to find interactions of a conformal field theory. The premise of the OPE is that the product of two conformally invariant observables (within a correlation function) can be expressed as the convergent sum of other observables. This is best illustrated in the $d = 2$ case via a Laurent expansion:

$$A(z)B(w) = \sum_{n=-\infty}^N \frac{C_n(w)}{(z-w)^n}. \quad (3.6)$$

Where $C_n(w)$ is some other field centered at w , which we index using n . These are generally different fields, but are usually found to be either $A(w)$ or $B(w)$ themselves. N is the order of truncation of the series.

The OPE only makes sense inside of correlation functions. It allows us to decompose high point correlators down into lower point correlators until we arrive at 2 and 3-point functions which are exactly determinable. An example of an operator product expansion is found in the appendix C, for the free fermion. Since the OPE encodes interactions in CFT, solving a model requires us finding the entire operator spectrum which appears in the OPE. For Minimal models this is finite, which is why they are exactly solvable.

3.4 Conformal Algebra and Radial Quantization

As seen in the literature review the conditions for conformal invariance in the case $d = 2$ requires our coordinate space to be parametrized by holomorphic and anti-holomorphic functions, thereby forcing the local conformal algebra to be formally infinite. In a $d = 2$ space with euclidean signature ($t \rightarrow -i\tau$), the local conformal algebra corresponds to the infinite sets of locally holomorphic and anti-holomorphic functions on the complex plane which completely parametrize the space. We see that this infinite algebra has

within it a closed sub-algebra of $SL(2, \mathbb{C})$ which corresponds to the global conformal transformations[17, 1]:

$$l_{-1} = -\partial_z, \quad \text{Translations,} \quad (3.7)$$

$$l_0 = -z\partial_z, \quad \text{Dilations and Rotations,} \quad (3.8)$$

$$l_1 = -z^2\partial_z, \quad \text{SCTs.} \quad (3.9)$$

A particular choice of these holomorphic and anti-holomorphic coordinates is the so-called radial scheme, where time and space translations correspond to radial translations and rotations respectively:

$$z = R_0 e^{\frac{2\pi}{L}w}, \quad \bar{z} = R_0 e^{\frac{2\pi}{L}\bar{w}}. \quad (3.10)$$

Where z is the holomorphic coordinate, and the $\bar{z}$ is the anti-holomorphic coordinate. The quantized version of the local conformal symmetry algebra is its so-called central extension, also known as the Virasoro algebra, and a closed $SL(2, \mathbb{C})$ sub-algebra is found to correspond to the global conformal transforms:

$$L_{-1} = -\partial_z, \quad \text{Translations,} \quad (3.11)$$

$$L_0 = -z\partial_z, \quad \text{Dilations and Rotations,} \quad (3.12)$$

$$L_1 = -z^2\partial_z, \quad \text{SCTs.} \quad (3.13)$$

We can specify pure dilations and rotations by combinations of the L_0 and $\bar{L}_0$ operator, such that we can recover the Hamiltonian and angular momentum operators respectively:

$$H = L_0 + \bar{L}_0, \quad J = L_0 - \bar{L}_0. \quad (3.14)$$

The Witt and Virasoro Algebras are presented in the next section where they are used to derive important results about CFT spectra. Having re-introduced the fundamental aspects of CFT, we proceed now to examine the minimal model framework.

4 Minimal Models

Minimal models were first conjectured by Belavin, Polyakov and Zamolodchikov in their seminal 1984 paper [2] as a class of exactly solvable statistical models in $d = 2$. They arise out of the infinite dimensionality of the 2D conformal algebra providing an infinite number of constraints on a theory, allowing us to exactly solve them, and giving us exact determination of the critical exponents. We have seen previously how gelation is modeled

by the $c = 0$ percolation model, and we will see in this section how the regular framework established in [2] breaks down when describing it. For contrast, a full treatment of the Ising model is carried out using the regular minimal model framework in C. The primary references for much of this chapter are [22, 1].

The primary aim of this section is to recall important results about the Virasoro algebra, and Verma modules, then introduce and illustrate how the standard minimal model framework arising from regular conformal invariance is inadequate for a complete description of $c = 0$ polymer gelation.

4.1 Verma Modules and the Virasoro Algebra

In order to model polymer gelation at criticality, we must use conformal symmetry, which is encoded in the Virasoro algebra, to impose restrictions on the theory. The Virasoro generator generates the Virasoro algebra, and is the mode expansion of the energy-momentum tensor. Therefore each Virasoro generator corresponds to a term in the OPE of the energy-momentum tensor with a given operator [17]:

$$T(z) = \sum_{n \in \mathbb{Z}} z^{n-2} L_n, \quad \bar{T}(\bar{z}) = \sum_{n \in \mathbb{Z}} \bar{z}^{n-2} \bar{L}_n. \quad (4.1)$$

The Virasoro algebra forms the foundation for all of applied 2D CFT and we present it here [1, 17]:

$$[L_n, L_m] = (n - m)L_{n+m} + \frac{c}{12}n(n^2 - 1)\delta_{n+m,0}. \quad (4.2)$$

We can see that for $c = 0$ CFT the Virasoro algebra reduces back to the Witt algebra, which is the symmetry algebra for classical conformal invariance:

$$[L_n, L_m] = (n - m)L_{n+m}. \quad (4.3)$$

This tells us that such quantum effects present in the $c = 0$ theory never break its conformal symmetry.

The operator $L_0 + \bar{L}_0 = H$ corresponds to an energy, and indeed the Virasoro generators encode relevant physical properties of the theory such as excitation energies, and angular momenta [16]. The spectra of these physical properties is contained in a special sort of vector space called a Verma module. We can present important results about Verma modules. Recall the conformal algebra, presented in B, and its analogous treatment to the harmonic oscillator. We will here draw a similar analogy now. All analysis below will be carried out with respect to the holomorphic sector of the Virasoro algebra, with the exact same results holding for the anti-holomorphic sector.

We define a so-called 'highest weight state' $|h\rangle$ which is analogous to the harmonic oscillator ground state. It is defined such that it is an eigenstate of L_0 with eigenvalue h , annihilated under the action of the SCT generator $L_n, n > 0$, and is invariant under action of the translation generator L_{-1} [22]:

$$L_0 |h\rangle = h |h\rangle, \quad L_{-1} |h\rangle = |h\rangle, \quad L_1 |h\rangle = 0. \quad (4.4)$$

An important subtlety is that these three generators cannot be used to generate the rest of the infinity of Virasoro modes as they form a closed sub-algebra of the full Virasoro algebra. This will have important implications later on. Another foundational result which will be important going forward[22]:

$$[L_0, L_m] = -mL_m, \quad [L_0, L_{-m}] = mL_{-m}. \quad (4.5)$$

The state-annihilation and creation conditions established for L_1 and L_{-1} can be further generalized to $L_n, L_{-n}, \forall n > 0$, and [22]:

$$L_n |h\rangle = 0, \quad L_{-n} |h\rangle = n |h\rangle, \quad \forall n > 0. \quad (4.6)$$

These generalizations can be proved by induction in the annihilation case, and by consideration of the creation operators in 4.5 in the second case. This in fact means that repeated action of the creation operator will increase the conformal weight, thereby creating the appropriate excited states [22, 1]:

$$L_{-k_1} L_{-k_2} \dots L_{-k_n} |h\rangle = (k_1 + k_2 + \dots + k_n) |h\rangle. \quad (4.7)$$

This is an eigenstate of L_0 , as we can see [22, 1]:

$$L_0 L_{-k_1} L_{-k_2} \dots L_{-k_n} |h\rangle = (h + k_1 + k_2 + \dots + k_n) |h\rangle = h + N |h\rangle. \quad (4.8)$$

Where we call the quantity N the level of the conformal field. This is distinct from the conformal dimension. Indeed, we can see that an infinite tower of states, which are degenerate in their levels are in fact formed by repeat action of the Virasoro generators on a field with highest weight state $|h\rangle$. We list the first such levels of the tower in 4.1.

We again reiterate the fact that in general $L_{-n} \neq (L_{-1})^n$. The structure we have described above is a so-called Verma module denoted $V(c, h)$ (with anti-holomorphic counterpart $\bar{V}(c, \bar{h})$), and it is a sub-space of the full Hilbert space, and comprises an infinite set of states generated by the infinite tower of negative operator modes.

Note that while the central charge c is a constant of the over-arching theory, the conformal weight h and $\bar{h}$ are dependent on the fields being considered (i.e. fermionic,

Figure 2: First five levels of a Verma Module. Figure credit: [1]

N (Level)	p(N) (Multiplicity)	Possible states
0	1	$ h\rangle$
1	1	$L_{-1} h\rangle$
2	2	$L_{-1}^2 h\rangle, L_{-2} h\rangle$
3	3	$L_{-1}^3 h\rangle, L_{-1}L_{-2} h\rangle, L_{-3} h\rangle$
4	5	$L_{-1}^4 h\rangle, L_{-1}^2L_{-2} h\rangle, L_{-2}^2 h\rangle, L_{-1}L_{-3} h\rangle, L_{-4} h\rangle$

bosonic, spinor, vector, scalar, etc.). As such in order to generate the full Hilbert space, we must sum over all the distinct particles in the system, and so by extension all the fields with distinct conformal weights. Indeed, our full Hilbert space is defined as [16]:

$$\mathcal{H} = \sum_{h, \bar{h}} V(c, h) \otimes \bar{V}(c, \bar{h}) \quad (4.9)$$

We can define an inner product on our Hilbert space by first defining a notion of conjugation [1]:

$$L_{-n} |h\rangle^\dagger = \langle h| L_n, \quad (4.10)$$

which we call a dual state. The identities presented and derived in the above section all hold but with the sign in the subscript and all appearances thereof reversed. The inner-product of two states is therefore given by[1]:

$$\langle h| L_{l_1} \dots L_{l_n} L_{-k_n} \dots L_{-k_1} |h\rangle \quad (4.11)$$

4.2 Minimal Models

As mentioned previously, the minimal models are exactly solvable CFTs, characterized by finite number of fields occurring in their OPEs. The ensuing sections will focus primarily on the minimal model framework in the context of $c = 0$ polymer gelation, which belongs to the poorly understood class of non-unitary CFTs[8, 11, 13]. A full example calculation for the critical Ising model, a very well understood conformal field theory, is found in the appendix C. We introduce the minimal model framework in this sub section so that in the next section we can apply it to the $c = 0$ model for polymer gelation.

4.2.1 Physical and Null States

The Verma module $V(c, h)$ is a sub-space of the full Hilbert space of a theory with central charge c . We need to find all the possible values of h and $\bar{h}$ admitted by the $c = 0$ theory.

To do this, we impose so-called physicality constraints:

$$L_n |phys\rangle = 0, (n > 0) \quad (L_0 - 1) |phys\rangle = 0 \quad (4.12)$$

This is incredibly important as it allows us to define a special sort of physical state called a null state, $|\chi\rangle$. This null state is orthogonal to the entire Hilbert space including itself, so has 0 norm, as well as satisfying all the conditions for a physical state[28]:

$$\langle\chi|phys\rangle = \langle phys|\chi\rangle^\dagger = 0, \forall |phys\rangle \quad (4.13)$$

Indeed, any two states differing by a null state are defined to be equivalent:

$$|phys\rangle \equiv |phys\rangle + |\chi\rangle \quad (4.14)$$

This null state is indicative of a redundant degree of freedom in our theory, and are sometimes known as gauge redundancies, or gauge symmetries. Imposing such redundancies on our theory comes into direct use in the following sections.

In order to restrict the number of fields in our CFT, we must impose that there exists a null state at a particular level in the Verma module. We list here the first 3 descendant states of the Verma module with their physical and null state conditions for the $c = 0$ polymer gelation model we are considering.

$$L_{-1} |h\rangle = |X_1\rangle, L_1 |X_1\rangle = 0, \langle X_1|X_1\rangle = 0, \quad (4.15)$$

$$\left(L_{-1}^2 - \frac{2}{3}L_{-2}\right) |h\rangle = |X_2\rangle, L_1 |X_2\rangle = L_2 |X_2\rangle = 0, \langle X_2|X_2\rangle = 0, \quad (4.16)$$

$$(L_{-1}^3 - 6L_{-2}L_{-1} + 12L_{-3}) |h\rangle = |X_3\rangle, L_1 |X_3\rangle = L_2 |X_3\rangle = L_3 |X_3\rangle = 0, \langle X_3|X_3\rangle = 0. \quad (4.17)$$

Where α, β, γ are all coefficients, which can be determined via the physicality and null-vector constraints. At each level we can collect all the linear combinations of operators inside a Gram matrix:

$$M^{(1)} = (\langle h| [L_1, L_{-1}] |h\rangle), \quad M^{(2)} = \begin{pmatrix} \langle h| [L_1^2, L_{-1}^2] |h\rangle & \langle h| [L_1^2, L_{-2}] |h\rangle \\ \langle h| [L_2, L_{-1}^2] |h\rangle & \langle h| [L_2, L_{-2}] |h\rangle \end{pmatrix},$$

$$M^{(3)} = \begin{pmatrix} \langle h| [L_1^3, L_{-1}^3] |h\rangle & \langle h| [L_1^3, L_{-2}L_{-1}] |h\rangle & \langle h| [L_1^3, L_{-3}] |h\rangle \\ \langle h| [L_1L_2, L_{-1}^3] |h\rangle & \langle h| [L_1L_2, L_{-2}L_{-1}] |h\rangle & \langle h| [L_1L_2, L_{-3}] |h\rangle \\ \langle h| [L_3, L_{-1}^3] |h\rangle & \langle h| [L_3, L_{-2}L_{-1}] |h\rangle & \langle h| [L_3, L_{-3}] |h\rangle \end{pmatrix} \dots \quad (4.18)$$

The explicit form of these matrices is found in terms of the conformal weight h and central charge c by computing the appropriate Virasoro commutators at each element. One can then arrive at polynomials in terms of h and c by setting the determinant of these matrices equal to 0, such as for the $c = 0$ case:

$$\det(M^{(1)}) = 2h = 0, \quad (4.19)$$

$$\det(M^{(2)}) = 2(16h^3 - 10h^2) = 0 = 32(h - h_{1,1})(h - h_{1,2})(h - h_{2,1}), \quad (4.20)$$

$$\begin{aligned} \det(M^{(3)}) &= 18h^6 + 24h^5 + 359h^4 + 296h^3 = 0 \\ &= A_3(h - h_{1,1})^2(h - h_{1,2})(h - h_{2,1})(h - h_{1,3})(h - h_{3,1})\dots \end{aligned} \quad (4.21)$$

Indeed, the determinant polynomial seen on the RHS of the expression is found to be a specific case of an object called the Kac-determinant with general expression[22, 1]:

$$\det(M^{(N)}) = A_N \prod_{r,s,rs \leq N} (h - h_{r,s})^{P(N-rs)}, \quad (4.22)$$

where the function $P(N-rs)$ denotes the partitions of the quantity $N-rs$ where $r, s \in \mathbb{Z}^+$, and A_N is some constant. By convention we must define $P(0) = 1$, as the only partition of 0 is itself: 0. Finding the zeroes of this determinant allows us to determine the number of distinct fields that exist in this theory $\phi_{r,s}$, and their scaling dimension $h_{r,s}$. In the level 1 case, we can see that $h = 0$, which corresponds to the so-called identity field. For the level 2 case, we can arrive at fields $\phi_{1,1}, \phi_{1,2}, \phi_{2,1}$, with respective scaling dimensions:

$$\begin{cases} h_{1,1} = 0, \\ h_{1,2} = \frac{1}{16} \left[(c-5) - \sqrt{(25-c)(1-c)} \right], \\ h_{2,1} = \frac{1}{16} \left[(c-5) + \sqrt{(25-c)(1-c)} \right]. \end{cases} \quad (4.23)$$

Where the association of $h_{1,2}$ with the negative root, and $h_{2,1}$ with the positive root are convention. There exist different yet equivalent standard parameterizations of $h_{r,s}$ and c . We will introduce here two specific parameterizations which will be useful for our treatment of percolation, though there exist others, which can be seen in [22, 1].

For coulomb gas models (models where the 2-point correlator that has an inverse square distance dependence), the following parametrization is found to be useful[22]:

$$h_{r,s} = h_0 + \frac{1}{4}(r\alpha_+ + s\alpha_-)^2, \quad (4.24)$$

$$h_0 = \frac{1}{24}(1-c), \quad (4.25)$$

$$\alpha = \frac{\sqrt{1-c}\sqrt{25-c}}{\sqrt{24}} \quad (4.26)$$

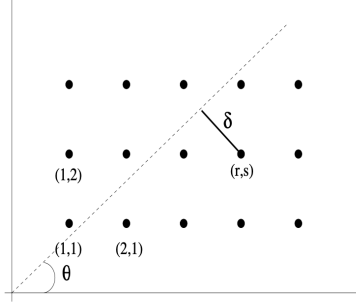


Figure 3: Kac-table. Points on the Kac-table correspond to primary fields of a theory.
Image Credit: [1]

Where the introduced parameters $\alpha_{+/-}$ are called charges. Additionally, we can find another parametrization in terms of different external parameter q , and $p = q + 1$:

$$c = 1 - \frac{6}{q(q+1)}, \quad (4.27)$$

$$h_{r,s} = \frac{[(q+1)r - qs]^2 - 1}{4q(q+1)}, \quad (4.28)$$

$$q = \frac{-1}{2} + / - \frac{1}{2} \sqrt{\frac{25-c}{1-c}} \quad (4.29)$$

We can see that both cases have defined the conformal weight $h_{r,s}$ in terms of these integers r, s which multiply to obtain the level of the null vector in the Verma module associated with the primary field $\phi_{r,s}$. These numbers r, s tell us the location of the field in a discretized theory space which we call the Kac-table, and the product $rs = N$ corresponds to the level at which a null-vector exists in the Verma module built from the primary field $\phi_{r,s}$. An example of the Kac-table is found in figure 3. We can interpret the Kac-table geometrically by introducing another parametrization of the Kac-determinant zeros[22]:

$$h_{r,s} = h_0 + \frac{1}{4}(r\alpha_+ + s\alpha_-)^2 \quad (4.30)$$

$$h_0 = \frac{1}{24}(c-1) \quad (4.31)$$

$$\alpha_{+/-} = \frac{\sqrt{1-c} + / - \sqrt{25-c}}{\sqrt{24}} \quad (4.32)$$

We call the quantities $\alpha_{+/-}$ charges.

The dashed line in Figure 1. has gradient: $\tan(\theta) = \frac{-\alpha_+}{\alpha_-}$. If the dashed line has irrational gradient, it will never meet any of the points in the Kac-table, but if it is rational, then there must exist two co-prime integers p , and q where $p > q$ such that[22]:

$$p\alpha_- + q\alpha_+ = 0 \Rightarrow \tan(\theta) = \frac{p}{q} \quad (4.33)$$

This then requires that the dashed line pass through point (q, p) . The zeros of the Kac-determinant therefore satisfy:

$$h_{r,s} = h_{r+q,s+p} = h_{q-r,p-s} \quad (4.34)$$

The line with gradient $-\frac{\alpha_+}{\alpha_-}$ is the so-called rationality constraint on the minimal model, and in the vernacular of CFT is the conservation of charge. The pair of integers (p, q) indicate the bounds of our theory space. This notion of bounding our theory space will be explored more rigorously in the proceeding section.

4.2.2 Differential Equations and Correlation Functions

Correlators are also known as propagators and in the regular QFT framework they tell us how fields evolve in space-time. It therefore is intuitive that they should satisfy the same physicality constraints as the underlying fields. This allows us to promote our null state requirements from operators on fields to homogeneous differential equations acting on correlation functions. We consider the primary field with conformal weight $h_{r,s}$, which we now assert is a null vector: $\phi_{r,s}^{null}$, which therefore has general expression:

$$\phi_{r,s}^{null}(z) = [\alpha_1 L_{-1}^{rs} + \alpha_2 L_{-1}^{rs-2} L_{-2} + \dots + \alpha_{rs} L_{rs}] \phi_{r,s}(z), \quad (4.35)$$

The coefficients α_i can be found entirely by imposing linear independence of all the vectors involved above. We can now see that any correlation function which has our null-vector above, must vanish, as Null states are orthogonal to all states of the Verma Modules including themselves (by definition they have 0 norm):

$$\langle \phi_{r,s}^{null}(z) \phi_1(z_1) \dots \phi_n(z_n) \rangle = 0 \quad (4.36)$$

To derive the differential equations, we now consider the level 2 descendant state:

$$|X_2\rangle = [L_{-2} - \frac{3}{2(2h+1)} L_{-1}^2] |h\rangle = L_{-2} \phi(z) - \frac{3}{2h+1} L_{-1}^2 \phi(z) \quad (4.37)$$

We will now prove an important result, which is that all correlation functions of descendant fields are obtained by acting with linear differential operators $\mathcal{L}_a$ on the correlators of the primary fields. We will see that the operators $\mathcal{L}_a$ are uniquely determined by the Virasoro Algebra.

We can prove this result through the definition of the Virasoro operator and the Ward identity. Consider the correlation function of primary fields $\phi_i(z_i), i = 1 \dots n-1$ with the

descendant field $L_{-k}\phi_n(z_n)$:

$$\begin{aligned}
\langle \phi_1(z_1) \dots \phi_{n-1}(z_{n-1}) (L_{-k}\phi_n)(z_n) \rangle &= \langle \phi_1(z_1) \dots \phi_{n-1}(z_{n-1}) T(z) \phi_n(z_n) \rangle \\
&= \frac{1}{2\pi i} \sum_{j=1}^{n-1} \oint_{z_j} dz (z - z_n)^{1-k} \left[\sum_{i=1}^n \left(\frac{h_i}{(z - z_i)^2} + \frac{1}{(z - z_i)} \partial_{z_i} \right) \right] \langle \phi_1 \dots \phi_n \rangle \\
&= - \sum_{j=1}^{n-1} \left[\frac{(1-k)h_j}{(z_j - z_n)^k} + \frac{1}{(z_j - z_n)^{k-1}} \partial_{z_j} \right] \langle \phi_1(z_1) \dots \phi_n(z_n) \rangle \\
&= \mathcal{L}_{-k} \langle \phi_1(z_1) \dots \phi_n(z_n) \rangle \square
\end{aligned}$$

Where in the first step we have used the definition of the Virasoro generator as the mode expansion of the stress-energy tensor, in the second line, we have introduced the operator product expansion of the stress-energy tensor with primary fields, seen in [1]. Indeed, by order of the Null-vector itself being expressible via linear combinations of states, we can therefore see that the correlator with the Null-vector inserted must satisfy:

$$\mathcal{D}_{rs} \langle \phi_{r,s}(z) \phi_1(z_1) \dots \phi_n(z_n) \rangle = 0 \quad (4.38)$$

Where $\mathcal{D}_{rs}$ is a linear differential operator of order rs .

The null vectors of a conformal family are in general infinite. For example there is a null vector at the level $(q-r)(p-s)$, which implies that the correlation function of $\phi_{r,s}(z)$ is not only the solution to the differential equation above, but is also the solution to the differential equation of order $\mathcal{D}_{(q-r)(p-s)}$. As such, a more general relation is Identified:

$$\mathcal{D}_n \langle \phi_{r,s}(z) \phi_1(z_1) \dots \phi_n(z_n) \rangle = 0 \quad (4.39)$$

Where n in the operator subscript is the level of the associated Null vector, and the order of the corresponding differential operator.

Indeed this now allows us to define an equivalence class which will allow us to implement a boundary of our Kac-table, by seeing that fields at or beyond the boundaries satisfy ill-defined differential equations. We begin with our previous definition of a minimal model:

$$p\alpha_- + q\alpha_+ = 0, \quad (4.40)$$

such that p , and q are co-prime. We now define $p = r+r'$, $q = s+s'$, such that $r, r', s, s' \in \mathbb{Z}$. We consider two fields $\phi_{r,s}$ and $\phi_{r',s'}$ to be equivalent such that:

$$r\alpha_- + s\alpha_+ = r'\alpha_- + s'\alpha_+. \quad (4.41)$$

We can see that this poses boundary conditions on the Kac-table, as any field with a $rs = n = 0$ has a level 0 null vector implying that the field itself is 0-valued. Additionally,

any fields outside the Kac-table can either be associated with those inside the Kac-table, or with other operators outside the Kac-table.

This notion of equivalence is what allows us to close the fusion ring of the minimal model, thereby enforcing a finite operator spectrum. Additionally, it can be seen that for certain combinations of r, s , the order n of the differential operator $\mathcal{D}_n$, that for $n < 0$, the differential operators become integral ones and are less well defined, leading us to exclude them from the theory space.

4.2.3 Fusion Rules and the OPE

In the above section, the correlator of a null-state with primary states was shown to satisfy a linear differential equation. By linearity, we know that a linear ODE's solutions can be made of linear combinations of other solutions. Indeed, in the above section we outlined that there are potentially several null-states of generally differing levels which can satisfy the same null-state condition, and correlator ODE, and it would be expedient to be able to generate these without having to solve such differential equations.

Consider the conformal field $\phi_{1,2}$. This conformal field has its first null-vector at level 2, which is given by:

$$\phi_{1,2}^{null}(z) = [L_{-2} - \frac{3}{2(2h_{1,2} + 1)}L_{-1}^2]\phi_{1,2}(z) \quad (4.42)$$

The correlator of this field with all other fields must therefore satisfy the differential equation:

$$\left[\frac{3}{2(2h_{1,2} + 1)}\partial_z^2 - \sum_{i=1}^n \left(\frac{h_i}{(z - z_i)^2} + \frac{1}{z - z_i}\partial_{z_i} \right) \right] \langle \phi_{1,2}(z)\phi_1(z_1)\dots\phi_n(z_n) \rangle = 0 \quad (4.43)$$

By considering the 2-point correlator of $\phi_{1,2}$ with another conformal field ϕ_h which has conformal weight h , then not only will the correlator satisfy the above ODE, but the OPE of $\phi_{1,2}$ and ϕ_h will as well:

$$\phi_{1,2}(z)\phi_h(z_1) = \sum_{h'} (C_{1,2,h})^{h'} (z - z_1)^{h' - h - h_{1,2}} \phi_{h'}(z_1) \quad (4.44)$$

Insertion of the RHS of 4.44 into the 2-point case of 4.43 yields:

$$\begin{aligned} \frac{3}{2(2h_{1,2} + 1)}(h' - h - h_{1,2})(h' - h - h_{1,2} - 1) - h + (h' - h - h_{1,2}) &= 0 \\ \text{Let } x = (h' - h - h_{1,2}) & \\ \Rightarrow \frac{3x(x - 1)}{2(2h_{1,2} + 1)} - h + x &= 0 \end{aligned} \quad (4.45)$$

This equation is second order in h' and therefore implies that there are two possible verma modules which can satisfy the requirement for a null-vector at the level $N = rs = 2$. Null-states imply that the Kac determinant is equal to 0, which in turn requires $h = h_{r,s}$, allowing us to solve for h' :

$$h' = \{h_{r,s-1}, h_{r,s+1}\} \quad (4.46)$$

Indeed, we can compactify the notation:

$$\phi_{1,2} \times \phi_{r,s} = [\phi_{r,s-1}] + [\phi_{r,s+1}]. \quad (4.47)$$

By performing a similar analysis for $\phi_{2,1}$ we get:

$$\phi_{2,1} \times \phi_{r,s} = [\phi_{r-1,s}] + [\phi_{r+1,s}]. \quad (4.48)$$

We can generalize these formulae by induction to:

$$\phi_{r,s} \times \phi_{r',s'} = \sum_{r''=|r-r'|+1}^{r+r'-1} \sum_{s''=|s-s'|+1}^{s+s'-1} \phi_{r'',s''}. \quad (4.49)$$

Where the summations in the above formula increment by 2 rather than 1. Indeed:

$$\begin{aligned} \phi_{1,2} \times \phi_{1,2} &= [\mathbb{I}] + [\phi_{1,3}] \\ \phi_{2,1} \times \phi_{2,1} &= [\mathbb{I}] + [\phi_{3,1}] \\ \phi_{1,2} \times \phi_{2,1} &= [\phi_{0,2}] + [\phi_{2,2}] = [\phi_{2,2}] \\ \phi_{2,1} \times \phi_{1,2} &= [\phi_{2,0}] + [\phi_{2,2}] = [\phi_{2,2}] \end{aligned}$$

In the last 2 equations above, we require that the OPE $\phi_{2,0}$ and $\phi_{0,2}$ vanish, which we call a truncation of the OPE, and implies that only finite number of fields occur in the OPE. This tells us that for a given equation of order rs , we can find all the null-states satisfying it by simply computing the OPE of the solutions. More broadly, these Fusion Rules allow us to move between Verma modules of a theory by changing the level at which there are zeros of the Kac determinant. We are at last on our way to constraining the number of Verma modules allowed by a theory.

4.3 $c = 0$, $\mathcal{M}_{2,3}$ and Percolation

Consider 4.27. We have shown in previous sections that polymer gelation corresponds to $c = 0$ percolation, and this has the result of simplifying the Virasoro algebra so that all central extensions vanish, giving rise to the Witt algebra. It also truncates our Kac-table

0	0	$\frac{1}{3}$	1	2	$\frac{10}{3}$	5	7	$\frac{28}{3}$	12	...
$\frac{5}{8}$	$\frac{1}{8}$	$-\frac{1}{24}$	$\frac{1}{8}$	$\frac{5}{8}$	$\frac{35}{24}$	$\frac{21}{8}$	$\frac{33}{8}$	$\frac{143}{24}$	$\frac{65}{8}$	...
2	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	1	2	$\frac{10}{3}$	5	...
$\frac{33}{8}$	$\frac{21}{8}$	$\frac{35}{24}$	$\frac{5}{8}$	$\frac{1}{8}$	$-\frac{1}{24}$	$\frac{1}{8}$	$\frac{5}{8}$	$\frac{35}{24}$	$\frac{21}{8}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Figure 4: The extended Kac-table for $c = 0$ model $\mathcal{M}_{2,3}$. The number in each square represents the associated conformal weight. The orientation is different from the previous figure. Image Credit [11]

theory space to $(2, 3)$:

$$c = 0 = 1 - \frac{6}{pq} \Rightarrow pq = 6, q = 2, p = 3, \quad (4.50)$$

$$h_{r,s} = \frac{(3r - 2s)^2 - 1}{24} \quad (4.51)$$

This means that the only possible fields which we can have in our theory are $\phi_{1,1}$ and $\phi_{1,2}$. All other fields must vanish as they are at or beyond the boundary of the Kac-table. Indeed, we can list the full set of field equivalencies for $\mathcal{M}_{2,3}$, beginning with $2\alpha_+ + 3\alpha_- = 0$:

$$1\alpha_+ + 1\alpha_- = 1\alpha_+ + 2\alpha_- \quad (4.52)$$

$$2\alpha_+ + 1\alpha_- = 0\alpha_+ + 2\alpha_- \quad (4.53)$$

$$1\alpha_+ + 3\alpha_- = 1\alpha_+ + 0\alpha_- \quad (4.54)$$

$$2\alpha_+ + 3\alpha_- = 0\alpha_+ + 0\alpha_- \quad (4.55)$$

We can also compute the extended Kac-table, which tells us the Kac-table beyond the boundary of the theory space. We present it here in figure 4.

4.3.1 Fusion Rules

Fusion tells us that interactions between two observables are entirely described in terms of another observable. Calculating these fusion rules tells us exactly the operator content of the theory, and thereby describes all the interactions of the theory. We can now compute the fusion rules for polymer gelation, using the equivalencies listed above, and formula 4.49:

$$\phi_{1,1} \times \phi_{1,1} = \phi_{1,1} \quad (4.56)$$

$$\phi_{1,2} \times \phi_{1,1} = \phi_{1,2} \quad (4.57)$$

$$\phi_{1,1} \times \phi_{1,2} = \phi_{1,2} \quad (4.58)$$

This predicts that the two fields which are used to model polymer gelation processes are the identity field $\phi_{1,1}$, and the field $\phi_{1,2}$ which has been found in the literature to indicate the start of a cluster. This is alarming as, having already identified the observable $\phi_{2,1}$ as a specifically desirable observable for our operator spectrum, it does not appear anywhere within the closed operator spectrum. The main reason for this is the association of the operator $\phi_{2,1}$ at the Kac-table boundary with fields outside the theory, which is intimately tied to the structure of the Virasoro algebra and the differential operator representations thereof acting on correlators. One possible way of circumventing this limitation has been to consider the logarithmic extension, which we shall now explore in the proceeding chapter.

5 Logarithmic Corrections: $\mathcal{LM}_{2,3}$ and $c = 0$

Having seen that the minimal model framework for rational CFT is inadequate for a complete description of polymer gelation, we now introduce the logarithmic extension to try and fix these problems. In this section, we show that even under logarithmic extension, the regular Virasoro algebra, as well as the notion of rationality in CFT, is wholly inadequate for finding non-trivial closed fusion rules for the $c = 0$ model of polymer gelation. We begin by introducing logarithmic CFT (LCFT), as a generalization of the regular CFT framework, and then progress to generalizing the idea of fusion to the LCFT framework. This will allow us to compute results in the latter two sections of this chapter demonstrating the existence of logarithmic behavior in the $c = 0$ LCFT, and also showing that the operator spectrum of the $c = 0$ polymer gelation model is infinite.

5.1 Jordan Block structure and Logarithmic Operators

Logarithmic extension arises out of generalizing the action of the Virasoro 0-mode from diagonal to non-diagonal. This will lead to logarithmic dependencies in our correlation functions, which is where the formalism derives its name. We primarily draw the framework from [11], and [13].

We begin by extending the action of the Virasoro L_0 mode to being non-diagonal:

$$L_0 |\Phi\rangle = h |\Phi\rangle \tag{5.1}$$

$$L_0 |\Psi\rangle = h |\Psi\rangle + |\Phi\rangle \tag{5.2}$$

Where the field $|\Phi\rangle$ is the logarithmic partner of the field $|\Psi\rangle$. We assume that the previously stated physicality condition $L_n |\Psi\rangle = 0$, ($n > 0$). We can collect the action of

the operator into a rank one Jordan block matrix which we can see is non-diagonalisable:

$$L_0 = \begin{pmatrix} h & 1 \\ 0 & h \end{pmatrix} \quad (5.3)$$

This action can be further generalized into an n -rank Jordan block matrix, though we do not present it here. Recall that the conformally invariant 2-point correlation function between two bosonic primary fields of the same conformal weight h has a fixed form:

$$\langle \phi(z) \phi(w) \rangle = \frac{C_{12}}{(z-w)^{2h}} \quad (5.4)$$

Where C_{12} is a proportionality constant fixed by the normalization of the inner product $C_{12} = \langle \phi | \phi \rangle$. For a field $|\Phi\rangle$, under the action of a rank 1 non-diagonal Virasoro operators with associated logarithmic partner field $|\Psi\rangle$, we arrive at:

$$C_{12} = \langle \Phi | \Phi \rangle = \langle \Phi | L_0 - h | \Psi \rangle. \quad (5.5)$$

Using $L_0 |\Phi\rangle = h |\Phi\rangle$.

To place constraints on correlation functions in the logarithmic formalism, we must identify new differential equations which will constrain the behavior of the correlators under symmetry transforms. We recall the differential representation of the Virasoro algebra used in the minimal models above:

$$\mathcal{L}_{-k} = \left[\frac{(1-k)h}{(z-w)^k} - \frac{1}{(z-w)^{k-1}} \partial_w \right] \quad (5.6)$$

It is desirable to alter this operator, and the field it acts upon in order to reproduce a non-diagonal action, which is done by including an extra Grassmann-valued term which accounts for an internal degree of freedom for our field, which will index the regular conformal primary, as well as its logarithmic partners[13]:

$$\Psi_{(h;n)}(z, \theta) = \sum_n a_n \frac{\theta^n}{n!} \Phi(z), \quad \mathcal{L}_{-k} = \left[\frac{(1-k)h}{(z-w)^k} - \frac{1}{(z-w)^{k-1}} \partial_w - \frac{\lambda(1-k)}{(z-w)^k} \partial_\theta \right] \quad (5.7)$$

Where a_n is an expansion coefficient, Φ is the regular conformal primary field from which logarithmic partners are generated, and θ is a Grassmann variable such that $\theta^n = 1$. For simplicity we consider a minimally logarithmic system, whereby a primary field $\Psi_{(h;0)} = \Psi$ is taken to have just one logarithmic partner $\Psi_{(h;1)} = \Phi$. Additionally, the inclusion of this extra internal degree of freedom to account for the logarithmic behavior is not well understood with respect to the notions of equivalence introduced previously, and is a topic of active research.

5.1.1 Logarithmic Correlators

To see where the formalism derives its name, we must place constraints on the correlation functions. As such, we now consider the group action of the global conformal generators on the partner field Φ . This is computed by considering that these commutators are implicitly composed with the conformally invariant ground state $|0\rangle$, and the definition of the Virasoro generator as the mode expansion of the energy-momentum tensor is been used:

$$[L_n, \Phi(w)] |0\rangle = L_n \Phi(w) |0\rangle = \oint_w \frac{dz}{2\pi i} T(z) \Phi(w) z^{n+1} \quad (5.8)$$

This integral is explicitly computed by recalling the divergent terms of the OPE of the energy-momentum tensor with the field, which correspond to the action of L_0 , and L_{-1} on the field, as the action of L_n , $n > 0$ must vanish by physicality constraints:

$$\oint_w \frac{dz}{2\pi i} T(z) \Phi(w) z \sim \oint_w \frac{dz}{2\pi i} \left[\frac{h\Phi + \Psi}{(z-w)^2} + \frac{\partial\Phi}{(z-w)} \right] \quad (5.9)$$

We can then correspond the appropriate expansion terms with the OPE.

$$[L_{-1}, \Psi] = \partial_z \Psi, \quad (5.10)$$

$$[L_0, \Psi] = h\Psi + \Phi + \partial_z \Psi \quad (5.11)$$

$$[L_1, \Psi] = 2hz\Psi + z^2\partial_z\Psi + 2z\Phi \quad (5.12)$$

Where in order to compute the final term we used the fact that $L_{-1} = w^2\partial_w = wL_0$. We can then use the arguments for the minimal models to upgrade these differential operators from acting on states to acting on 2-point functions, yielding a set of inhomogeneous differential equations which the two-point operators satisfy:

$$\begin{aligned} (\partial_z + \partial_w) \langle \Psi(z) \Phi(w) \rangle &= 0, \\ (z\partial_z + w\partial_w + 2h) \langle \Psi(z) \Phi(w) \rangle &= -\langle \Psi(z) \Psi(w) \rangle, \\ (z^2\partial_z + w^2\partial_w + 2h(z+w)) \langle \Psi(z) \Phi(w) \rangle &= -2w \langle \Psi(z) \Psi(w) \rangle. \end{aligned}$$

$$\begin{aligned} ((z-w)(\partial_z + \partial_w) + 2h) \langle \Phi(z) \Phi(w) \rangle &= -2 \langle \Phi(z) \Psi(w) \rangle, \\ (z\partial_z + w\partial_w + 2h) \langle \Phi(z) \Phi(w) \rangle &= -2 \langle \Phi(z) \Psi(w) \rangle, \\ (z^2\partial_z + w^2\partial_w + 2h(z+w)) \langle \Phi(z) \Phi(w) \rangle &= -2(z+w) \langle \Phi(z) \Psi(w) \rangle. \end{aligned}$$

We consider that this system of differential equations admits general solutions:

$$\langle \Phi(z) \Phi(w) \rangle = 0, \quad \langle \Phi(z) \Psi(w) \rangle = \frac{B}{(z-w)^{2h}}, \quad \langle \Psi(z) \Psi(w) \rangle = \frac{C - 2B \log(z-w)}{(z-w)^{2h}} \quad (5.13)$$

An important addendum is that it is possible that there are regular primaries which do not produce logarithmic partners, which can coexist in a theory with primary fields which do generate logarithmic partners. To distinguish between the two we will refer to the latter from now on as 'pre-logarithmic' primary fields.

5.2 Generalizing Fusion

Fusion has so-far been considered as a particular property of the operator product expansion, and has therefore implicitly been working as an operation on the primary fields on top of which the Verma modules are built. We now abandon this notion in it's entirety, and instead work on the full Verma module. By generalizing fusion, we will be able to identify when operators exhibit regular, and logarithmic conformal invariance in the spectrum.

Without loss of generality, we can quotient out the sub-modules, without losing anything of import. We now consider 4 objects [11]:

1. $\mathcal{V}_n$ is the Verma module with a null-vector at level $n = rs$.
2. $Q_{r,s} = \mathcal{V}_0/V_n, n = rs$, is the quotient module, and reduces the Verma module from an infinite to a finite dimensional space. This will not affect much of the prescriptive treatment, but is a necessary distinction to make for implementation of the NGK algorithm, particularly in numerical settings.
3. $\mathcal{L}_h$ is the irreducible representation of this quotient space, and it is seen as permissible to set $\mathcal{L}_{h_{r,s}} = Q_{r,s}$, i.e. the Quotient space is in it's irreducible representation.
4. $\mathcal{S}_{r,s}$ is a non-highest weight module. This object contains the non-diagonal behavior of L_0 which leads to logarithmic corrections, and is defined as such:

$$\mathcal{S}_{1,3k-1} = Q_{1,3k-1} \oplus Q_{1,3k-5}, \quad \mathcal{S}_{1,3k-2} = Q_{1,3k-2} \oplus Q_{1,3k-4}, \quad k \in \mathbb{Z}^+. \quad (5.14)$$

This direct sum of the two quotient spaces can be thought of as the container of the 'proper' primary field and it's logarithmic partner field. It is convention in LCFT to assume the logarithmic partner field to be of a lower level to the 'proper' primary field.

There exists a prescriptive method for computing fusion rules, which is underpinned by the Nahm-Gabardiel-Kausch (NGK) algorithm. We present the NGK Algorithm as well as it's prescriptive counterpart to verify some elementary results, and show how logarithmic behaviour manifests itself in $c = 0$.

5.2.1 The Nahm-Gabardiel-Kausch Algorithm

The Nahm Gabardiel Kausch (NGK) algorithm is a more general way of computing fusion rules in conformal field theory which accounts for rational and logarithmic behaviour. We introduce it primarily following [11]. As mentioned previously, we now upgrade the fusion product to a tensor product $\otimes$ as the maximal combination of the two vector spaces. This means that the resultant space built from M and N dimensional Verma modules is now an $|M \times N|$ dimensional vector space, which we call the fusion module, and in order for it to satisfy the appropriate null-vector conditions, we must therefore generalize the action of the Virasoro modes by use of the co-product formulae:

$$\Delta(L_n) = \sum_{m=-1}^n \binom{n+1}{m+1} L_m \otimes \mathbb{I} + \mathbb{I} \otimes L_n, \quad (n \geq -1), \quad (5.15)$$

$$\Delta(L_{-n}) = \sum_{m=-1}^{\infty} (-1)^{m+1} \binom{n+m-1}{m+1} L_m \otimes \mathbb{I} + \mathbb{I} \otimes L_{-n}, \quad (n \geq 2) \quad (5.16)$$

$$L_{-n} \otimes \mathbb{I} = \sum_{m=n}^{\infty} \Delta(L_{-m}) + (-1)^n \sum_{m=-1}^{\infty} \binom{n+m-1}{m+1} \mathbb{I} \otimes L_m, \quad (n \geq 2). \quad (5.17)$$

Where $\binom{k}{l} = \frac{k!}{l!(k-l)!}$. A generalization of this formula exists for other symmetry algebras [11, 26, 13], though it is beyond the scope of this thesis. The definition above of the quotient module allows us to truncate the infinite Verma modules into a finite dimensional subspaces, without losing the logarithmic behavior which we are interested in.

In addition, it is not strictly necessary to consider the entirety of the resulting subspace, as we can define a cut-off level g , beyond-which any vector generated by Virasoro modes is simply set equal to 0:

$$L_{-n_1} L_{-n_2} \dots L_{-n_k} |h\rangle = 0, \quad \text{where, } (n_1 + n_2 + \dots + n_k \geq g) \quad (5.18)$$

This truncation renders all the co-product formulae summations finite, and therefore tractable for our purposes. To complete the fusion calculation, we must now find the null states in the new tensor product space. We denote the fusion module with the cut-off as $(M \times N)^{(g)}$.

We can prove by induction that when the fusion product is the direct product of two modules given the co-products we have defined (for now we do not specify any further structure inside the modules), then we can define the equivalence [11]:

$$(M \times N)^{(g)} = M' \otimes N^{(g)} \quad (5.19)$$

Where M' is the special sub-space such that $L_{-n_1} \dots L_{-n_k} |h\rangle = 0$ when $\max\{n_1, \dots, n_k\} > 1$. M' is the subspace created by the sole action of L_0, L_{-1} (all positive mode operators anni-

hilate), and $N^{(g)}$ is the regular cut-off module. We display the algorithm in an incremental fashion:

1. Compute the basis vectors of the component vector spaces of $(M \times N)^{(g)} = M' \otimes N^{(g)}$.
2. Compute the co-product of the Virasoro operators associated with the basis Virasoro modes.
3. Compute the action of the co-product operators on the basis vectors of the space $(M \times N)^{(g)}$, and compute the eigenvectors. If there are degenerate eigenvalues, this implies the existence of a Jordan block structure in $\Delta(L_0)$.
4. Increment the cut-off by 1.

5.3 Logarithmic behavior in Polymer Gelation

Algorithms are best demonstrated by use-cases. We here show the existence of regular and logarithmic behavior in polymer gelation fields via the NGK algorithm via the fusion of modules $Q_{1,2}$, with $Q_{1,2}$ and latterly $Q_{1,2}$ with $\mathcal{L}_{1/3}$, following closely arguments in [11]

5.3.1 $Q_{1,2} \times Q_{1,2}$:

We consider the $g = 0$ cut-off: $(Q_{1,2} \times Q_{1,2})^{(g)} = Q'_{1,2} \times Q_{1,2}^{(0)}$. At $g = 0$, the full vector space is spanned by:

$$\begin{cases} |h\rangle \otimes |h\rangle \\ L_{-1} |h\rangle \otimes |h\rangle \end{cases}$$

The action of the co-product $\Delta(L_0)$ yields:

$$\begin{aligned} \Delta(L_0) |h\rangle \otimes |h\rangle &= L_{-1} |h\rangle \otimes |h\rangle \\ \Delta(L_0) L_{-1} |h\rangle \otimes |h\rangle &= \frac{1}{3} L_{-1} |h\rangle \otimes |h\rangle \end{aligned}$$

This tells us that $\Delta(L_0)$ is both diagonalizable and has eigenvalues $h_{1,1} = 0$, $h_{1,3} = 1/3$. This tells us the conformal weights of the associated highest weight states. In order to identify the location of singular vectors in the fusion product, we can now compute to grade $g = 1$. To perform this calculation we enforce the $\Delta(L_{-1})^2 = \Delta(L_{-2}) = 0$, which is the level 2 null state condition which is found in the $Q_{1,2}$ module. As such, we impose:

$$\begin{aligned} 0 &= \Delta(L_{-1})^2(|h\rangle \otimes |h\rangle) = L_{-1}^2 |h\rangle \otimes |h\rangle + 2L_{-1} |h\rangle \otimes L_{-1} |h\rangle + |h\rangle \otimes L_{-1}^2 |h\rangle \\ &\Rightarrow 0 = \frac{2}{3} L_{-1} |h\rangle \otimes |h\rangle + 2L_{-1} |h\rangle \otimes L_{-1} |h\rangle - \frac{2}{3} |h\rangle \otimes L_{-1} |h\rangle \\ &\Rightarrow 2L_{-1} |h\rangle \otimes L_{-1} |h\rangle = 0 \quad (5.20) \end{aligned}$$

This tells us that at grade 1 the singular vector occurs at level 1, as $\Delta(L_{-1})$ annihilates the $h = 0$ state. The highest weight module is therefore $Q_{1,1}$. Further calculation will tell us that $Q_{1,3} = \mathcal{L}_{1/3}$ is in it's irreducible representation[11].

The resulting fusion product is:

$$Q_{1,2} \times Q_{1,2} = Q_{1,1} \oplus \mathcal{L}_{1/3} \quad (5.21)$$

Nowhere in our calculation have we encountered non-diagonal operators. We proceed to compute another fusion product, which we hope will exhibit logarithmic behavior.

5.3.2 $Q_{1,2} \times \mathcal{L}_{1/3}$:

We now consider the fusion product $Q_{1,2} \times \mathcal{L}_{1/3}$. We begin with the grade 0 calculation, where the vector space is spanned by $|h_{1,2}\rangle \otimes |h_{1,3}\rangle$ and $L_{-1} |h_{1,2}\rangle \otimes |h_{1,3}\rangle$. The co-products $\Delta(L_0)$ and $\Delta(L_{-1})$ are given by:

$$\Delta(L_0) = (L_{-1} \otimes \mathbb{I}) + 2(\mathbb{I} \otimes L_0) + (L_0 \otimes \mathbb{I}) \quad (5.22)$$

$$\Delta(L_{-1}) = (L_{-1} \otimes \mathbb{I}) + (\mathbb{I} \otimes L_{-1}) \quad (5.23)$$

Computation to grade 0 yields the eigenvalues 0 and 1. We now extend the grade to $g = 1$, and compute the action of $\Delta(L_0)$:

$$\Delta(L_0)(|h_{1,2}\rangle \otimes |h_{1,3}\rangle) = L_{-1} |h_{1,2}\rangle \otimes |h_{1,3}\rangle \quad (5.24)$$

$$\Delta(L_0)(L_{-1} |h_{1,2}\rangle \otimes |h_{1,3}\rangle) = L_{-1} |h_{1,2}\rangle \otimes |h_{1,3}\rangle \quad (5.25)$$

$$\Delta(L_0)(|h_{1,2}\rangle \otimes L_{-1} |h_{1,3}\rangle) = |h_{1,2}\rangle \otimes L_{-1} |h_{1,3}\rangle \quad (5.26)$$

$$\Delta(L_0)(L_{-1} |h_{1,2}\rangle \otimes L_{-1} |h_{1,3}\rangle) = 2L_{-1} |h_{1,2}\rangle \otimes L_{-1} |h_{1,3}\rangle \quad (5.27)$$

The co product $\Delta(L_0)$ has eigenvalues: $(0, 1, 1, 2)$ and so is not necessarily diagonalisable implying the logarithmic structure we would hope the system to exhibit.

At both grades the action of $\Delta(L_0)$ on $|h_{1,2}\rangle \otimes |h_{1,3}\rangle$ has produced a state with eigenvalue 0. As such, we define $|\phi_{1,2}\rangle \equiv |h_{1,2}\rangle \otimes |h_{1,3}\rangle$. We consider that both of the basis vectors of eigenvalue 1 can be associated with $|\phi_{1,4}\rangle$. Indeed, we also see that the action:

$$\Delta(L_{-1}) |\phi_{1,2}\rangle = L_{-1} |h_{1,2}\rangle \otimes |h_{1,3}\rangle + |h_{1,2}\rangle \otimes L_{-1} |h_{1,3}\rangle \quad (5.28)$$

Which we can see are the two states with eigenvalue 1 corresponding to $\Delta(L_0)$ in the basis.

We can multiply by -1 , and re-arrange this into a suggestive form:

$$\Delta(L_0) (L_{-1} |h_{1,2}\rangle \otimes |h_{1,3}\rangle) = -|h_{1,2}\rangle \otimes L_{-1} |h_{1,3}\rangle + \Delta(L_{-1}) |\phi_{1,2}\rangle. \quad (5.29)$$

To collect this into a Jordan block, we realize that it must also hold true that:

$$\Delta(L_0)\Delta(L_{-1})|\phi_{1,2}\rangle = \Delta(L_{-1})|\phi_{1,2}\rangle \quad (5.30)$$

This condition yields $L_{-1}|h_{1,2}\rangle \otimes |h_{1,3}\rangle \propto -|h_{1,2}\rangle \otimes L_{-1}|h_{1,3}\rangle$, which can then be normalized such that $L_{-1}|h_{1,2}\rangle \otimes |h_{1,3}\rangle = -|h_{1,2}\rangle \otimes L_{-1}|h_{1,3}\rangle$. This allows us to use an altered Jordan block format for the action of $\Delta(L_0)$ on $|\phi_{1,4}\rangle = L_{-1}|h_{1,2}\rangle \otimes |h_{1,3}\rangle = -|h_{1,2}\rangle \otimes L_{-1}|h_{1,3}\rangle$:

$$\Delta(L_0) = \begin{pmatrix} h_{1,4} & 1 \\ 0 & h_{1,4} \end{pmatrix} \Rightarrow \begin{cases} \Delta(L_0)|\phi_{1,4}\rangle = |\phi_{1,4}\rangle + \Delta(L_{-1})|\phi_{1,2}\rangle \\ \Delta(L_0)\Delta(L_{-1})|\phi_{1,2}\rangle = \Delta(L_{-1})|\phi_{1,2}\rangle \end{cases} \quad (5.31)$$

Which we can see clearly tells us $|\phi_{1,4}\rangle$ has logarithmic partner $\Delta(L_{-1})|\phi_{1,2}\rangle$, which implies the existence of logarithmic divergences. The eigenvalue $h = 2$ state is found to be the generated by action of $\Delta(L_{-1})$ on the logarithmic partner field $\Delta(L_{-1})|\phi_{1,2}\rangle$.

We can see that the states we are interested in necessarily have a regular primary field with $h = 0$, and a Jordan block for $h = 1$. Reading these off the Kac-table, we can associate them with the quotient modules $Q_{1,2}$ and $Q_{1,4}$ respectively, and we contain these modules inside the module $\mathcal{S}_{1,4} = Q_{1,2} \oplus Q_{1,4}$.

Finally, this gives us the fusion rule:

$$Q_{1,2} \times \mathcal{L}_{1/3} = \mathcal{S}_{1,4}. \quad (5.32)$$

As such we have seen that logarithmic behavior appears naturally in polymer gelation.

5.4 The Operator Spectrum

We here attempt a naive derivation of logarithmic fusion rules for $\mathcal{LM}_{3,2}$, using a prescriptive method that we will now introduce, which will allow us to prove that the operator spectrum of polymer gelation is infinite. The calculations in the above sections are tedious beyond $g = 1$, and are often undertaken with computer algebra packages [11]. Much of the rigorous calculation has been left out in favor of heuristic approaches in the interest of space, and appropriate justifications can be found in [21, 11]. However, there does exist a prescriptive method for computing logarithmic fusion rules which matches exactly the computed data for the NGK algorithm[11].

1. For a spectrum consisting of irreducible modules $Q_{1,3k}$ and logarithmic modules $\mathcal{S}_{1,3k-1}, \mathcal{S}_{1,3k-2}$, for $k \in \mathbb{Z}^+$, we must break the logarithmic modules down into the associated partner

fields:

$$\mathcal{S}_{1,3k-1} \rightarrow Q_{1,3k-1} \oplus Q_{1,3k-5} \quad \mathcal{S}_{1,3k-2} \rightarrow Q_{1,3k-2} \oplus Q_{1,3k-4}. \quad (5.33)$$

2. Then compute the fusion via the distributive property, and the prescription:

$$Q_{1,s} \times Q_{1,s'} = \bigoplus_{s''=|s-s'|+1}^{s+s'+1} Q_{1,s''} \quad (5.34)$$

Where the direct summation increments by 2 each step.

3. Re-compose the $\mathcal{S}_{1,3k-1}$ and $\mathcal{S}_{1,3k-2}$ modules in a unique way.

Indeed this prescriptive method can be generalized via induction to:

$$Q_{r,s} \times Q_{r',s'} = \bigoplus_{s''=|s-s'|+1}^{s+s'+1} \bigoplus_{r''=|r-r'|+1}^{r+r'+1} Q_{r'',s''}. \quad (5.35)$$

Where the summation increments by 2 in each index. This prescriptive method implies an infinite operator spectrum, which we will prove by contradiction. We impose the condition that the operator spectrum is closed on it's upper bound:

$$\begin{aligned} s' + s - 1 &< s' \\ \Rightarrow s &< 1, s' < s' \square \end{aligned}$$

We can see that this admits a singular solution $s = 0$, and the argument holds equally for r , leading to closure iff $r = 0$, which we recall is also trivial by the Kac-boundary conditions, which still apply in the logarithmic case. By the boundary conditions of the Kac-table, this means the only way that the fusion product can close under composition with other fields in the Kac-table is by composition with the identity, implying that the operator spectrum generated by fusion is infinite.

6 Conclusions and Outlook

The goal of this report was to attempt to find a closed set of fusion rules which would allow us to exactly solve polymer gelation. We have failed to do so in both the rational and logarithmic frameworks. However, this result is actually in good agreement with the literature, and more importantly numerical results. See [24] where the authors computed the bootstrap of a conformal field theory with vanishing central charge, and an infinite operator spectrum generated by non-diagonal Virasoro modes. Their results were in good agreement with Monte-Carlo simulations of percolation carried out by the same authors.

A promising potential avenue of further research is to attempt to extend the notions of notions of equivalency used in rational CFT to logarithmic CFT. This has so far been achieved through extensions of the underlying symmetry algebra. In [14] the authors extend the regular Virasoro Algebra to the so-called triplet algebra in order to solve the $c = -2$ model, associated with the self-avoiding random walk of critical dense polymers. This approach is then built on in [25] and [26] where they close percolation under the $\mathcal{W}$ -extension of the Virasoro Algebra. However, an understanding of $\mathcal{W}$ -algebras and their associated representation theory is presently outside the competencies of the author.

Further promising extensions in this direction could be within the framework of Super-Conformal Field Theory, and Logarithmic extension, taking $\mathcal{LM}_{2,3} \rightarrow \mathcal{LSM}_{2,3}$. This direction involves the extension of the Virasoro Algebra to the Virasoro Super-Algebra. For example the $\mathcal{N} = 1$ Virasoro Super-Algebra is given by:

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{n+m,0}, \quad (6.1)$$

$$[L_m, G_r] = \left(\frac{m}{2} - r\right)G_{m+r}, \quad (6.2)$$

$$\{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0} \quad (6.3)$$

Indeed, a treatment of such extensions is given in [23], and a further study is found in [5] and [6]. A treatment with appropriate background in polymers is found in [27], which considers the $\mathcal{N} = 2$ extended super-symmetry algebra.

All possible further research in this area however works on the assumption that we expect percolation to exhibit, or has imposed upon it, extra symmetries beyond simply conformal ones at criticality. This is not clear to be true to the author apriori. Indeed, logarithmic CFT is viewed as a generalization of CFT to a non-diagonal action of the Virasoro Algebra on certain primary fields. As such it is not clear that there isn't some further, more general framework of which logarithmic CFT is itself a subset. Therefore, a no-go theorem in a similar spirit to the Haag-Sohnius-Lopuszański and Coleman-Mandula theorems, which restricts the imposition of further symmetries on statistical systems at criticality and their corresponding CFTs, or even simply on critical percolation would therefore be very useful indeed.

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A The Percolation Phase Transition

Consider the percolation partition function of the $Q \rightarrow 1$ -state Potts model:

$$Z_{perc} = \lim_{Q \rightarrow 1} (Z_{Potts}) = \lim_{Q \rightarrow 1} \sum_{\mathcal{C}} p^{|\mathcal{C}|} (1-p)^{B-|\mathcal{C}|} Q^{N(\mathcal{C})} = \sum_{\mathcal{C}} p^{|\mathcal{C}|} (1-p)^{B-|\mathcal{C}|} \quad (\text{A.1})$$

Where we sum over the connected components in the graph. To show that a phase transition exists, we consider the associated Gibbs free-energy:

$$F = -\ln(Z_{perc}) = -\ln \left(\sum_{\mathcal{C}} p^{|\mathcal{C}|} (1-p)^{B-|\mathcal{C}|} \right) \quad (\text{A.2})$$

up to a constant $k_b T$, which we normalize to 1. To interrogate the system in the vicinity of criticality, we take $p = p_c + \delta p$, where p_c is the critical bond existence probability. This means we look at the limit where $\delta p \rightarrow 0$ from above and below. For a phase transition to exist we must show that there exists a discontinuity within derivatives of the Gibbs free energy with respect to p in the region of criticality. To first order:

$$\left. \frac{\partial F}{\partial(\delta p)} \right|_{\delta p \rightarrow 0} = \frac{-1}{Z_{perc}} \frac{\partial Z_{perc}}{\partial(\delta p)} = \sum_{\mathcal{C}} [|\mathcal{C}|(\delta p)^{-1} - (1-\delta p)^{-1}(B-|\mathcal{C}|)] \quad (\text{A.3})$$

We can see that the result appears to have both divergent and convergent terms as $\delta p \rightarrow 0$, corresponding to the first and second terms respectively. However, since we are looking at percolation in the bulk, without any consideration for boundary conditions, we can take an additional limit on the cluster size $|\mathcal{C}| \rightarrow \infty$ such that $|\mathcal{C}| \propto (\delta p)^{-1}$. This tells us that in the first term, $|\mathcal{C}| \rightarrow \infty$ at a rate which will be proportional to $(\delta p)^{-1} \rightarrow \infty$, meaning we can expect the divergences to cancel. As such, we consider the second order derivative:

$$\left. \frac{\partial^2 F}{\partial(\delta p)^2} \right|_{\delta p \rightarrow 0} = \sum_{\mathcal{C}} [-|\mathcal{C}|(\delta p)^{-2} + (1-\delta p)^{-2}(B-|\mathcal{C}|)], \quad (\text{A.4})$$

where we can see the divergence in the first term will not cancel, as we expect $(\delta p)^{-2} \rightarrow \infty$ faster than $|\mathcal{C}| \rightarrow \infty$. As we are looking at percolation in the bulk, we also examine the limit of the total cluster size $|\mathcal{C}| \rightarrow \infty$. Therefore, we can see that percolation is a second order phase transition, which we know are well described by CFTs. We can also directly derive the central charge of the theory:

$$\Delta F = -\Delta \ln(Z_{perc}) = -\Delta \ln(1) = 0 = -\frac{1}{6} c \chi \ln(L) \Big|_{L \rightarrow \infty} \implies c = 0 \square. \quad (\text{A.5})$$

A.1 Limits

The essence of the above argument relies on the assertion that $|\mathcal{C}| \rightarrow \infty$ at the same rate as $(\delta p)^{-1} \rightarrow \infty$. We now would like to prove this. The total cluster-size is related to the

clustering coefficient of its constituent nodes. As such, for a connected component with V' vertices:

$$|\mathcal{C}| = \sum_{i \in V'} k_i \langle C_i(A) \rangle \quad (\text{A.6})$$

where k_i is the number of edges incident on node i , and $\langle C_i \rangle$ is the ensemble average clustering coefficient, which measures the fraction of nodes which connect to nodes in the same connected component. Because we have taken an ensemble average, we may proceed without specifying any topological aspects of the lattice. For simple non-directed graphs, the ensemble average clustering coefficient is given by:

$$\langle C_i(A) \rangle = p - (pN - p^2)(1 - p)^{N-2}. \quad (\text{A.7})$$

In order to show that this quantity diverges at the same rate as p , we take a derivative. As such, we consider:

$$\left. \frac{\partial |\mathcal{C}|}{\partial p} \right|_{p \rightarrow 0,1} = \sum_{i \in V'} k_i \left. \frac{\partial}{\partial p} \langle C_i(A) \rangle \right|_{p \rightarrow 0,1}. \quad (\text{A.8})$$

$$\left. \frac{\partial}{\partial p} \langle C_i(A) \rangle \right|_{p \rightarrow 0,1} = 1 + (N - 2)(pN - p^2)(1 - p)^{N-3} - (N - 2p)(1 - p)^{N-2} \quad (\text{A.9})$$

$$\Rightarrow \left. \frac{\partial}{\partial p} \langle C_i(A) \rangle \right|_{p \rightarrow 1} = 1 \quad (\text{A.10})$$

$$\Rightarrow \left. \frac{\partial}{\partial p} \langle C_i(A) \rangle \right|_{p \rightarrow 0} = 1 - N \quad (\text{A.11})$$

$$\Rightarrow \left. \frac{\partial}{\partial p} \langle C_i(A) \rangle \right|_{p \rightarrow 0} = 1 \square. \quad (\text{A.12})$$

Where we have used the fact that as $p \rightarrow 0$, the size of connected components N must tend to 0 in the maximally disordered phase. Indeed, we have now proved that:

$$\left. \frac{\partial |\mathcal{C}|}{\partial p} \right|_{p \rightarrow 0} = \sum_{i \in V'} k_i \in \mathbb{Z}, \quad (\text{A.13})$$

meaning that $|\mathcal{C}|$ is some integer multiple of p , so we expect it to diverge at the same rate, thereby allowing us to make the above argument.

B 2D CFT

In the literature review component of this thesis, 2D CFT was introduced primarily using [1] and [17], and in a general fashion without motivation. We will now re-introduce the important aspects of CFT here with a focus on physical intuition, and by analogy with systems which are already well known to physicists.

B.1 Conformal Field Theory

We will primarily present important results about 2D CFT, that have already been derived as part of the literature review.

B.1.1 The Conformal Group and Algebra

Conformal transformations are mappings on a manifold which leave notions of angle invariant. We can see that this is an equivalent statement to the requirement that the metric is invariant up to a position-dependent re-scaling under coordinate transforms (diffeomorphisms):

Indeed, it can be seen in standard references [1, 17], and was shown in the literature review, that the resulting condition on the metric: $g_{\alpha\beta} = \Omega(x)g_{\mu\nu}$, yields the below constraints on the infinitesimal coordinate transform $x^\mu \rightarrow x'^\mu = x^\mu + \epsilon^\mu$:

$$f(x)g_{\mu\nu} = (\partial_\nu \epsilon_\mu + \partial_\mu \epsilon_\nu), \quad (\text{B.1})$$

$$f(x) = \frac{2}{d}(\partial_\nu \epsilon^\nu), \quad (\text{B.2})$$

$$2\partial_\mu \partial_\nu \epsilon_\rho = g_{\nu\rho} \partial_\mu f + g_{\rho\mu} \partial_\nu f - g_{\mu\nu} \partial_\rho f, \quad (\text{B.3})$$

$$2\partial^2 \epsilon_\mu = (2-d)\partial_\mu f, \quad (\text{B.4})$$

$$(2-d)\partial_\nu \partial_\mu f = g_{\mu\nu} \partial^2 f, \quad (\text{B.5})$$

$$(d-1)\partial^2 f = 0. \quad (\text{B.6})$$

Where we have introduced a 'normalized scale-factor' $f(x) = 1 - \Omega(x)$. The results B.1 - B.6 can be combined in such a way as to derive the set of finite global conformal transforms which comprise the conformal group:

$$\text{Translations:} \quad x'_\mu = x_\mu + a_\mu, \quad (\text{B.7})$$

$$\text{Rotations:} \quad x'_\mu = (\mathbb{I} + m_{\mu\nu})x_\mu, \quad (\text{B.8})$$

$$\text{Dilations:} \quad x'_\mu = (1 + \lambda)x_\mu, \quad (\text{B.9})$$

$$\text{Special Conformal Transformations:} \quad x'_\mu = x^\mu + 2(\vec{x} \cdot \vec{b})x^\mu - b^\mu x^2. \quad (\text{B.10})$$

We recognize that translations and rotations are members of the Poincaré group. Indeed, it is possible to recover the Poincaré group as a subgroup of the Conformal Group via the condition $\Omega(x) = 1$.

To understand the action of the conformal group on Quantum States, we must now move from the finite transforms to the infinitesimal generators of the Conformal Group. By considering $\delta\phi = \phi'(x) - \phi(x)$, and $x = x' - \delta x$, for $\delta \ll 1$ one can arrive at the appropriate generators:

$$\begin{aligned}
\text{Translation: } P_\mu &= -i\partial_\mu, \\
\text{Rotation: } M_{\mu\nu} &= i(x_\nu\partial_\mu - x_\mu\partial_\nu), \\
\text{Dilatation: } D &= -ix^\mu\partial_\mu, \\
\text{SCT: } K_\mu &= -i(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu).
\end{aligned}$$

Taking the commutator allows us to arrive at the Conformal Algebra:

$$[D, P_\mu] = iP_\mu, \quad (\text{B.11})$$

$$[D, K_\mu] = -iK_\mu, \quad (\text{B.12})$$

$$[D, M_{\mu\nu}] = 0, \quad (\text{B.13})$$

$$[K_\mu, K_\nu] = 0, \quad (\text{B.14})$$

$$[M_{\mu\nu}, K_\rho] = -i(\eta_{\nu\rho}K_\mu - \eta_{\mu\rho}K_\nu), \quad (\text{B.15})$$

$$[P_\mu, K_\nu] = 2i(M_{\mu\nu} - \eta_{\mu\nu}D). \quad (\text{B.16})$$

Where all other commutators vanish[1]. We can gain some intuition for what the conformal algebra actually means with regards to it's action on quantum mechanical systems by considering an analogy with the quantum harmonic oscillator:

$$\hat{H} = \hat{a}^\dagger\hat{a} + E_0, \quad \text{where we normalize } \hbar\omega = 1. \quad (\text{B.17})$$

If we take the commutator of $\hat{H}$ with $\hat{a}$ and $\hat{a}^\dagger$, we can see that it reproduces a similar commutator to and B.12:

$$\begin{aligned}
[\hat{H}, \hat{a}] &= [\hat{a}^\dagger\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}]\hat{a} + \hat{a}^\dagger[\hat{a}, \hat{a}] = -\mathbb{I}\hat{a} \sim [D, K_\mu], \\
[\hat{H}, \hat{a}^\dagger] &= [\hat{a}^\dagger\hat{a}, \hat{a}^\dagger] = [\hat{a}^\dagger, \hat{a}^\dagger]\hat{a} + \hat{a}^\dagger[\hat{a}, \hat{a}^\dagger] = \mathbb{I}\hat{a} \sim [D, P_\mu].
\end{aligned}$$

We can therefore make several associations with regards to the unique generators of the conformal group. The Dilation operator D fulfills the role of the Hamiltonian operator in conformally symmetric theories, whereas the SCT, and Translation operators now correspond to annihilation and creation operators respectively.

To extend this analogy, we define a so-called highest-weight state (the reasoning behind this name will become apparent later), which will play the role of the ground state in the conformal quantization scheme $|\phi\rangle = \phi(0)|0\rangle$. We specify the conformal group elements

to have the following actions:

$$D|\phi\rangle = i\Delta|\phi\rangle, \quad (\text{B.18})$$

$$K_\mu|\phi\rangle = 0, \quad (\text{B.19})$$

$$P_\mu|\phi\rangle = \Delta + 1|\phi\rangle, \quad (\text{B.20})$$

$$\frac{i}{2}\omega^{\mu\nu}M_{\mu\nu}|\phi\rangle = g_I^J|\phi_J\rangle \quad (\text{B.21})$$

Where in the last relation, the indices I and J indicate specific matrix representations of the Lorentz group. The Lorentz Rotation is of marginal importance in 2D CFTs, as we will see later, so no treatment is given here. Of particular note in the above is the action of P_μ on the highest weight state. We can see that it acts as a ladder operator, and this will become important when treating the Virasoro Algebra later. The eigenvalue Δ is the so-called scaling dimension of the field, and a more thorough treatment of this is found in the below section.

B.1.2 Correlation Functions

In regular QFT, correlation functions are the transition amplitudes between fields at different spacetime points, and are therefore subject to the same equations of motion as the fields. This leads to very difficult to solve differential equations in order to calculate these correlation functions. However in CFT the imposition of Special Conformal and Dilational symmetries, forces enough conditions to exactly determine both 2 and 3 point functions up to a constant factor. Under rotations and translations: $x^\mu \rightarrow x^\mu + a^\mu$, and $x^\mu \rightarrow \Lambda_\nu^\mu x^\nu$ requires that the correlation function be dependent on the magnitude of distance:

$$\langle\phi(x_1)\phi(x_2)\rangle = f(|x_1 - x_2|). \quad (\text{B.22})$$

We expect the correlation function to transform as the products of two quasi-primary fields as in 3.2, which we will now take to have different scaling dimensions in general, so under dilatation $x^\mu \rightarrow x'^\mu = \lambda x^\mu$:

$$\begin{aligned} \langle\phi_1(x_1)\phi_2(x_2)\rangle &\rightarrow \lambda^{\Delta_1+\Delta_2}\langle\phi_1(x_1)\phi_2(x_2)\rangle = f(\lambda|x_1 - x_2|) = \langle\phi_1(x_1)\phi_2(x_2)\rangle, \\ &\Rightarrow \lambda^{\Delta_1+\Delta_2}f(|x_1 - x_2|) = f(\lambda|x_1 - x_2|), \\ &\Rightarrow f(|x_1 - x_2|) = \frac{C_{12}}{|x_1 - x_2|^{\Delta_1+\Delta_2}} = \langle\phi_1(x_1)\phi_2(x_2)\rangle. \end{aligned} \quad (\text{B.23})$$

Special conformal invariance requires that the fields in our correlator have the same scaling dimension, otherwise it will vanish, and we can show this given the special conformal

transform and an important transformation property:

$$\gamma_i \equiv 1 - 2b \cdot x_i + b^2 x_i^2 = (1 - b \cdot x_i)^2, \quad |x'_i - x'_j| = \frac{|x_i - x_j|}{\gamma_i^{1/2} \gamma_j^{1/2}}. \quad (\text{B.24})$$

Where we use the special conformal transform $x^\mu \rightarrow x'^\mu = \gamma_i^{1/2} x^\mu$. This tells us that under the equivalent transform $x^\mu \rightarrow \gamma_i^{1/2} x^\mu$ the correlation function will transform as:

$$\begin{aligned} \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}} &= \frac{1}{\gamma_1^{\Delta_1/2} \gamma_2^{\Delta_2/2}} \frac{C_{12}}{|x'_1 - x'_2|^{\Delta_1 + \Delta_2}} = \frac{(\gamma_1 \gamma_2)^{(\Delta_1 + \Delta_2)/2}}{\gamma_1^{\Delta_1/2} \gamma_2^{\Delta_2/2}} \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}}, \\ &\Rightarrow \begin{cases} C_{12} = 0, \\ \Delta_1 = \Delta_2. \end{cases} \end{aligned}$$

The first case corresponds to the trivial case where the correlator vanishes in it's entirety, the second case allows us to recover the original correlator from conformal invariance. This allows us to arrive at the final version of the 2-point function:

$$\langle \phi_1(x_1) \phi_2(x_2) \rangle = \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}}, \quad \begin{cases} C_{12} = 0, \\ \Delta_1 = \Delta_2. \end{cases} \quad (\text{B.25})$$

Similar arguments can be followed in the 3-point case to restrict them to the form:

$$\langle \phi_1(x_1) \phi_2(x_2) \phi_3(x_3) \rangle = \frac{C_{123}}{x_{12}^{\Delta_1 + \Delta_2 - \Delta_3} x_{23}^{\Delta_2 + \Delta_3 - \Delta_1} x_{13}^{\Delta_3 + \Delta_1 - \Delta_2}}. \quad (\text{B.26})$$

Where $x_{ij} = |x_i - x_j|$.

B.2 The Case d=2

Conformal field theory becomes profoundly rich in 2D, largely as a result of the 2D local conformal group being infinite dimensional. This richness leads to models becoming exactly solvable. To begin our treatment, we consider the constraints B.1-B.6. The Killing equation is obtained by combining B.1 and B.2, [12]:

$$(\partial_\nu \epsilon_\mu + \partial_\mu \epsilon_\nu) = \frac{2}{d} (\partial_\nu \epsilon^\nu) g_{\mu\nu}. \quad (\text{B.27})$$

Where we recall that d corresponds to space-time dimensions. Component-wise inspection and imposing $d = 2$ yields the Cauchy-Riemann relations of complex analysis:

$$\partial_0 \epsilon_0 + \partial_0 \epsilon_0 = \partial_1 \epsilon_1 + \partial_0 \epsilon_0, \quad \Rightarrow \quad \partial_0 \epsilon_0 = \partial_1 \epsilon_1, \quad (\text{B.28})$$

$$\partial_0 \epsilon_1 + \partial_1 \epsilon_0 = 0, \quad \Rightarrow \quad \partial_0 \epsilon_1 = -\partial_1 \epsilon_0. \quad (\text{B.29})$$

Therefore the 2D conformal transformation ϵ_μ must be either a holomorphic or anti-holomorphic function. Indeed the space can be parametrized in terms of holomorphic and anti-holomorphic functions [12, 1]:

$$z = x^0 + ix^1 \quad \epsilon = \epsilon^0 + i\epsilon^1 \quad \partial_z = \frac{1}{2}(\partial_0 - i\partial_1), \quad (\text{B.30})$$

$$\bar{z} = x^0 - ix^1 \quad \bar{\epsilon} = \epsilon^0 - i\epsilon^1 \quad \partial_{\bar{z}} = \frac{1}{2}(\partial_0 + i\partial_1). \quad (\text{B.31})$$

Such that the Cauchy-Riemann relations simplify to:

$$\partial_z \bar{\epsilon} = 0, \quad \partial_{\bar{z}} \epsilon = 0. \quad (\text{B.32})$$

This has several striking implications. The first of which being that we now re-parametrize fields in a conformally symmetric space in terms of the new variables $z, \bar{z}$ as follows:

$$\phi(x^\mu) \rightarrow \phi(z, \bar{z}) \quad (\text{B.33})$$

Our Primary field transformation rule 3.2 now then generalizes to:

$$\phi(z, \bar{z}) \rightarrow \left(\frac{\partial z'}{\partial z} \right)^{-h} \left(\frac{\partial \bar{z}'}{\partial \bar{z}} \right)^{-\bar{h}} \phi(z, \bar{z}) \quad (\text{B.34})$$

Where we have introduced the so-called "conformal weights" h , and $\bar{h}$. These play a similar role to the scaling dimension introduced in the previous section, and follow the relation $\Delta = h + \bar{h}$. A subtle point here is that $\bar{h} \neq h^*$, where $*$ is the usual complex conjugation operation.

This shift into complex coordinates is interpreted as being equivalent to performing a wick rotation, thereby moving from a Minkowski metric to a Euclidean metric. While it is possible to continue in $d = 2$ using a minkowski metric and entirely real ϵ^μ , as is done in string theory, we will continue with the euclidean coordinates specified above.

B.2.1 The Local Conformal Group

We may show that the local conformal group is indeed infinite dimensional, by considering that as holomorphic and anti-holomorphic functions, the infinitesimal conformal transforms $\varepsilon^\mu(z)$ and $\bar{\varepsilon}(\bar{z})$ admit Laurent series expansion about the origin:

$$z' = z + \varepsilon(z), \quad \varepsilon(z) = \sum_{n=-\infty}^{\infty} c_n z^{n+1}, \quad (\text{B.35})$$

$$\bar{z}' = \bar{z} + \bar{\varepsilon}(\bar{z}), \quad \bar{\varepsilon}(\bar{z}) = \sum_{n=-\infty}^{\infty} \bar{c}_n \bar{z}^{n+1}. \quad (\text{B.36})$$

Considering now the action of these transforms on the field $\phi(z, \bar{z})$, given that the CI condition $\phi'(z', \bar{z}') = \phi(z, \bar{z})$ is imposed:

$$\begin{aligned}
\phi(z', \bar{z}') &= \phi(z, \bar{z}) + \delta\phi(z, \bar{z}), \\
&\Rightarrow \phi(z + \varepsilon(z), \bar{z} + \varepsilon(\bar{z})) = \phi(z, \bar{z}) + \delta\phi(z, \bar{z}), \\
&\Rightarrow \phi(z, \bar{z}) + \delta\phi(z, \bar{z}) = \phi(z, \bar{z}) + \varepsilon(z)\partial_z\phi(z, \bar{z}) + \bar{\varepsilon}(\bar{z})\partial_{\bar{z}}\phi(z, \bar{z}) + O(\varepsilon^2, \bar{\varepsilon}^2), \\
&\Rightarrow \delta\phi(z, \bar{z}) = \sum_{n=-\infty}^{+\infty} -c_n z^{n+1} \partial_z \phi(z, \bar{z}) - \bar{c}_n \bar{z}^{n+1} \partial_{\bar{z}} \phi(z, \bar{z}), \\
&\Rightarrow \delta\phi(z, \bar{z}) = \sum_{n=-\infty}^{+\infty} c_n l_n \phi(z, \bar{z}) + \bar{c}_n \bar{l}_n \phi(z, \bar{z}). \quad (\text{B.37})
\end{aligned}$$

Where we have defined the generators:

$$l_n = -z^{n+1} \partial_z, \quad \bar{l}_n = -\bar{z}^{n+1} \partial_{\bar{z}}. \quad (\text{B.38})$$

Because there are an infinite number of 2D holomorphic and anti-holomorphic functions, the local conformal group in 2D has an infinite set of the above such generators which form the Witt algebra:

$$[l_n, l_m] = (n - m)l_{n+m}, \quad (\text{B.39})$$

$$[\bar{l}_n, \bar{l}_m] = (n - m)\bar{l}_{n+m}, \quad (\text{B.40})$$

$$[l_n, \bar{l}_m] = 0, \quad m, n \in [-\infty, \infty]. \quad (\text{B.41})$$

This algebra is infinite dimensional, and particular cases of these generators correspond to global conformal symmetries:

$$l_{-1} = -\partial_z \quad \text{Generates Translations in } \mathbb{C}, \quad (\text{B.42})$$

$$l_0 = -z\partial_z \quad \text{Generates Dilations and Rotations in } \mathbb{C}, \quad (\text{B.43})$$

$$l_1 = -z^2\partial_z \quad \text{Generates SCTs in } \mathbb{C}. \quad (\text{B.44})$$

The quantized version of Witt algebra is the Virasoro algebra, it upgrades conformal symmetry of a geometric space to symmetry of a Hilbert space by adding a so-called central extension[18, 12, 3]:

$$[L_n, L_m] = (n - m)L_{n+m} + \frac{c}{12}n(n^2 - 1)\delta_{n+m,0}, \quad (\text{B.45})$$

$$[\bar{L}_n, \bar{L}_m] = (n - m)\bar{L}_{n+m} + \frac{c}{12}n(n^2 - 1)\delta_{n+m,0}, \quad (\text{B.46})$$

$$[L_n, \bar{L}_m] = 0. \quad (\text{B.47})$$

The Virasoro modes are defined as the mode expansion of the energy-momentum tensor, and we can see that these modes correspond to the same generators as the Witt algebra in the same cases:

$$L_{-1} = -\partial_z. \quad \text{Translations.} \quad (\text{B.48})$$

$$L_0 = -z\partial_z. \quad \text{Dilations and rotations.} \quad (\text{B.49})$$

$$L_1 = -z^2\partial_z. \quad \text{SCTs.} \quad (\text{B.50})$$

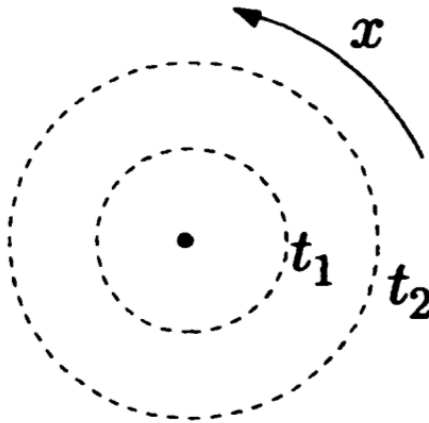
B.2.2 Radial Quantization and the OPE

We have seen above that by analogy the dilatation operator D corresponds to the Hamiltonian operator in conformally symmetric theories. We can now make this analogy more explicit by making a specific choice for $z, \bar{z}$ [4]:

$$z = R_0 e^{\frac{2\pi}{L}w}, \quad \bar{z} = R_0 e^{\frac{2\pi}{L}\bar{w}}. \quad (\text{B.51})$$

Where $w = \tau + ix$, and $\bar{w} = \tau - ix$. R_0 is some characteristic length scale which can be ignored. $\tau = -it$ corresponds to euclidean time arising from the Wick rotation discussed previously. The map is conformal as $\partial_{\bar{w}}z(w) = \partial_w\bar{z}(\bar{w}) = 0$. This transforms our coordinate space such that it looks like the following: Where integrals over space at equal time

Figure 5: Radial Coordinate System, Image Credit: [1]



now correspond to circular contours, about a singularity (as $\tau \rightarrow -\infty, z \rightarrow 0$). This has two important corollaries. The first is that this choice of coordinates admits a notion of

radial ordering of fields which is equivalent to time ordering [1]:

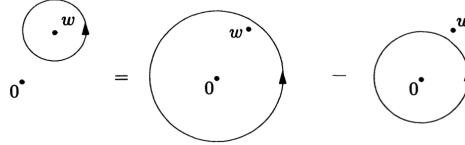
$$\mathcal{R}(\Phi_1(z)\Phi_2(w)) = \mathcal{T}(\Phi_1(z)\Phi_2(w)) = \begin{cases} \Phi_1(z)\Phi_2(w) & \text{if } |z| > |w|, \\ \Phi_2(w)\Phi_1(z) & \text{if } |z| < |w|. \end{cases} \quad (\text{B.52})$$

A minus sign is added in front of the second expression for fermionic fields to preserve their anti-commutation relations. The second corollary is that integrals over all space at equal times are now contour integrals about the origin. This allows us to formulate an alternate definition of the commutator for different :

$$[A(z), B(w)] = \oint_0 dw \oint_{C_w} dz a(z)b(w). \quad (\text{B.53})$$

Where C_w is a contour about the point w . This commutator definition can be used to derive the virasoro algebra from the mode expansion of the energy momentum tensor. We can graphically represent the above condition as:

Figure 6: Contour Integral Subtraction, Image Credit: [1]



We can now introduce the OPE, or Operator Product Expansion, which is a key construct in Conformal Field Theory. The OPE allows us to express products of fields inside a correlation function, (referred to as operators in the literature), located at different space-time points as a new operator located at one of those positions.

We remember that correlation functions must abide themselves abide by the symmetries of the physical theory they are a part of, and so also must the constituent fields. This is combined with the already established fact that 2D local conformal invariance requires functions be holomorphic, or anti-holomorphic, and in both cases imposes analyticity.

The Operator product expansion is an example of an Operator Algebra. We can conceptually liken the Operator Algebra here to spin-operators seen in non-relativistic quantum mechanics. Recall that the spin operators have the commutation relations:

$$[\hat{S}_x, \hat{S}_y] = i\hat{S}_z, \quad [\hat{S}_y, \hat{S}_z] = i\hat{S}_x, \quad [\hat{S}_x, \hat{S}_z] = i\hat{S}_y.$$

Here we can see that our commutator has taken two operators in and produced a third different operator. This is in principle what an Operator Algebra does. It is a multiplication rule for fields (operators), which allows us to express the product of two operators

as a third operator in the same basis. One can in fact generalize this notion from a single operator to a sum over a potential infinity of other local operators[22]:

$$\phi_p(z)\phi_q(w) = \sum_{r=0}^{\infty} \frac{C_{pq}^r}{|z-w|^{h_p+h_q-h_r}} \phi_r(w). \quad (\text{B.54})$$

Where C_{pq}^r are the structure constants of the algebra. We call B.54 the Operator Product Expansion.

C The Critical Ising Model

We here present a full treatment of the Ising model at criticality which corresponds to a $c = 1/2$ CFT, thereby lying within the framework of the minimal models.

C.1 $c = 1/2$ Free Majorana Fermion in $d = 2$

The association between the Ising model and the free fermion model is well known. We begin with the free fermion in $d = 2$, and must find its central charge before we can examine the minimal model. To do this, we must calculate the operator product expansion of the fermionic fields, and we take the fermions to be Majorana. In 2 dimensions the action of the free Majorana fermion is given by:

$$S = \frac{g}{2} \int d^2x \Psi^\dagger \gamma^0 \gamma^\mu \partial_\mu \Psi \quad (\text{C.1})$$

Where the Dirac matrices γ^μ satisfy the Clifford algebra:

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \quad (\text{C.2})$$

Indeed in the Euclidean formalism where $\eta_{\mu\nu} = \text{diag}(1, 1)$ there are only 2 dirac matrices:

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \gamma^1 = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (\text{C.3})$$

and the term $\gamma^0 \gamma^\mu \partial_\mu$ in our action becomes:

$$\gamma^0 (\gamma^0 \partial_0 + \gamma^1 \partial_1) = 2 \begin{pmatrix} \partial_{\bar{z}} & 0 \\ 0 & \partial_z \end{pmatrix} \quad (\text{C.4})$$

By writing our 2-component spinor as $\Psi = (\psi, \bar{\psi})$, meaning that the action becomes:

$$S = g \int d^2x (\bar{\psi} \partial \bar{\psi} + \psi \bar{\partial} \psi) \quad (\text{C.5})$$

We now have to calculate the 2-point correlator $\langle \Psi_i \Psi_j \rangle$, where $i, j = 1, 2$. To do this we re-express our action in a Gaussian form:

$$S = \frac{1}{2} \int d^2x d^2y \Psi_i A_{ij} \Psi_j$$

where $A_{ij} = g\delta(x-y)(\gamma^0\gamma^\mu)_{ij}\partial_\mu$

$$\Rightarrow A_{ij}A_{ij}^{-1} = \delta_{ij} = g\delta(x-y)(\gamma^0\gamma^\mu)_{ij}\partial_\mu A_{ij}^{-1}$$
(C.6)

We can again now solve for the 2-point greens function as before, although in this case we get a matrix of solutions:

$$2g \begin{pmatrix} \partial_{\bar{z}} & 0 \\ 0 & \partial_z \end{pmatrix} \begin{pmatrix} \langle \psi(z, \bar{z})\psi(w, \bar{w}) \rangle & \langle \psi(z, \bar{z})\bar{\psi}(w, \bar{w}) \rangle \\ \langle \bar{\psi}(z, \bar{z})\psi(w, \bar{w}) \rangle & \langle \bar{\psi}(z, \bar{z})\bar{\psi}(w, \bar{w}) \rangle \end{pmatrix} = \frac{1}{\pi} \begin{pmatrix} \partial_{\bar{z}} \frac{1}{z-w} & 0 \\ 0 & \partial_z \frac{1}{\bar{z}-\bar{w}} \end{pmatrix} \quad (C.7)$$

Where we have as before translated $x \rightarrow (z, \bar{z})$ and $y \rightarrow (w, \bar{w})$. We can now easily read off solutions:

$$\langle \psi(z, \bar{z})\psi(w, \bar{w}) \rangle = \frac{1}{2\pi g} \frac{1}{(z-w)} \quad (C.8)$$

$$\langle \bar{\psi}(z, \bar{z})\bar{\psi}(w, \bar{w}) \rangle = \frac{1}{2\pi g} \frac{1}{(\bar{z}-\bar{w})} \quad (C.9)$$

$$\langle \bar{\psi}(z, \bar{z})\psi(w, \bar{w}) \rangle = \langle \psi(z, \bar{z})\bar{\psi}(w, \bar{w}) \rangle = 0 \quad (C.10)$$

Which are the truncation terms of the Laurent series. As such we arrive at the OPE for the fermionic field as:

$$\psi(z)\psi(w) \sim \frac{1}{2\pi g} \frac{1}{(z-w)} \quad (C.11)$$

Indeed we can see that this obeys fermi-dirac statistics as by swapping $z \leftrightarrow w$, we can see that we pick up an extra - sign. To now compute the OPE of the energy-momentum tensor with the field ψ and itself, we can use the same expression as before, however we now list out the various components as the field itself has matrix structure:

$$T^{\bar{z}\bar{z}} = 2 \frac{\partial \mathcal{L}}{\partial(\bar{\partial}\Psi)} \partial\Psi = 2g\psi\partial\psi \quad (C.12)$$

$$T^{zz} = 2 \frac{\partial \mathcal{L}}{\partial(\partial\Psi)} \bar{\partial}\Psi = 2g\bar{\psi}\bar{\partial}\bar{\psi} \quad (C.13)$$

$$T^{z\bar{z}} = 2 \frac{\partial \mathcal{L}}{\partial(\partial\Psi)} \partial\Psi - 2\mathcal{L} = -2g\psi\bar{\partial}\psi \quad (C.14)$$

We can now compute the OPE using Wick's theorem of the Energy-momentum tensor with itself:

$$\begin{aligned}
T(z)\psi(w) &= -\pi g : \psi(z)\partial\psi(z) : \psi(w) \\
&\Rightarrow T(z)\psi(w) = \frac{1}{2} \frac{\partial\psi(z)}{(z-w)} + \frac{1}{2} \frac{\psi(z)}{(z-w)^2} \\
&\Rightarrow T(z)\psi(w) = \frac{\frac{1}{2}\psi(w)}{(z-w)^2} + \frac{\partial\psi(w)}{(z-w)}
\end{aligned} \tag{C.15}$$

And of the Energy-momentum tensor with itself:

$$T(z)T(w) \sim \frac{\frac{1}{4}}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{(z-w)} \implies c = \frac{1}{2} \tag{C.16}$$

C.2 Fusion Rules

We have now shown that the Ising Model corresponds to a $c = \frac{1}{2}$ conformal field theory, which is clearly rational. We know that it will have conformal weight given by:

$$\begin{aligned}
h_{r,s} &= h_0 + \frac{1}{4}(r\alpha_+ + s\alpha_-)^2 \\
\alpha_{+/-} &= \frac{\sqrt{1-c} + / - \sqrt{25-c}}{\sqrt{24}} = \frac{\sqrt{\frac{1}{2}} + / - \sqrt{24\frac{1}{2}}}{\sqrt{24}}
\end{aligned}$$

The gradient of the corresponding line in the Kac table is given by:

$$-\frac{\alpha_+}{\alpha_-} = -\frac{\sqrt{1-c} + \sqrt{25-c}}{\sqrt{1-c} - \sqrt{25-c}} \tag{C.17}$$

$$= -\frac{\sqrt{1/2} + \sqrt{49/2}}{\sqrt{1/2} - \sqrt{49/2}} \tag{C.18}$$

$$= -\frac{\sqrt{1/2} + 7\sqrt{1/2}}{\sqrt{1/2} - 7\sqrt{1/2}} \tag{C.19}$$

$$= -\frac{8}{-6} = \frac{4}{3} \tag{C.20}$$

$$\frac{\alpha_+}{\alpha_-} = -\frac{4}{3} \tag{C.21}$$

This tells us that the Minimal model is given by $\mathcal{M}_{4,3} = \mathcal{M}_{p,q}$. Indeed, we can use this to now derive some constraints:

$$\alpha_+ + \alpha_- = -(2\alpha_+ + 3\alpha_-) \Rightarrow \phi_{1,1} = \phi_{2,3} \quad (\text{C.22})$$

$$2\alpha_+ + \alpha_- = -(\alpha_+ + 3\alpha_-) \Rightarrow \phi_{2,1} = \phi_{1,3} \quad (\text{C.23})$$

$$\alpha_+ + 2\alpha_- = -(2\alpha_+ + 2\alpha_-) \Rightarrow \phi_{1,2} = \phi_{2,2} \quad (\text{C.24})$$

$$3\alpha_+ + 1\alpha_- = -(3\alpha_-) \Rightarrow \phi_{3,1} = \phi_{0,3} = 0 \quad (\text{C.25})$$

$$(\text{C.26})$$

Which implies that:

$$\begin{aligned} h_{1,1} &= h_0 + \frac{1}{4}(\alpha_+ + \alpha_-)^2 \\ &= h_0 + \frac{1}{4}(-(2\alpha_+ + 3\alpha_-))^2 \\ &= h_0 + \frac{1}{4}(2\alpha_+ + 3\alpha_-)^2 \\ &= h_{2,3} \\ h_{2,1} &= h_{1,3} \\ h_{1,2} &= h_{2,2} \end{aligned}$$

We can now choose to derive the set of fusion rules associated with this model:

$$\phi_{1,1} \times \phi_{1,1} = [\phi_{1,1}] \quad (\text{C.27})$$

$$\phi_{1,1} \times \phi_{1,2} = [\phi_{1,2}] \quad (\text{C.28})$$

$$\phi_{1,1} \times \phi_{2,1} = [\phi_{2,1}] \quad (\text{C.29})$$

$$(\text{C.30})$$

This intuitively makes sense: earlier we identified the Verma module with null vector at 1, 1 as the Identity. Then by the above constraints:

$$\phi_{1,2} \times \phi_{1,2} = [\phi_{1,1}] + [\phi_{2,1}] \quad (\text{C.31})$$

$$\phi_{2,1} \times \phi_{2,1} = [\phi_{1,1}] \quad (\text{C.32})$$

$$\phi_{1,2} \times \phi_{2,1} = [\phi_{2,2}] = [\phi_{1,2}] \quad (\text{C.33})$$

$$\phi_{2,1} \times \phi_{1,2} = [\phi_{2,2}] = [\phi_{1,2}] \quad (\text{C.34})$$

We can see that the algebra here is entirely closed. we have successfully shown that the minimal model $\mathcal{M}_{4,3}$ has its eigenvalue spectrum generated entirely by a finite number of Verma modules. A similar argument can be applied to other minimal models, with

2	1/2	1/16	0
1	0	1/16	1/2
$\mathcal{M}_{4,3}$	1	2	3

Table 2: Kac-Table of $\mathcal{M}_{4,3}$

Kac-Table indices	Conformal Weight	Primary Field	Observable
(1,1) and (3,2)	0	$\mathbb{I}$	Identity
(2,1) and (2,2)	1/2	ϵ	Energy
(1,2) and (3,1)	1/16	σ	Spin

Table 3: Primary fields and their observables. Credit [1, 22]

conformal weights given by:

$$h_{1,1} = 0, h_{1,2} = \frac{1}{16}, h_{2,1} = \frac{1}{2} \quad (\text{C.35})$$

A summation of the product of the holomorphic and anti-holomorphic Verma modules over the conformal weights, as in (1.10), the full hilbert space is recovered.

We gather this all together in the appropriate fusion and Kac-Tables2: We find the associated Ising Model scaling dimensions and the associated operators3: Symbolically, the fusion rules are [1, 22]:

$$\begin{aligned}
\mathbb{I} \times \mathbb{I} &= \mathbb{I} \\
\mathbb{I} \times \epsilon &= \epsilon \\
\mathbb{I} \times \sigma &= \sigma \\
\epsilon \times \epsilon &= \mathbb{I} \\
\sigma \times \sigma &= \mathbb{I} + \epsilon \\
\sigma \times \epsilon &= \sigma
\end{aligned}$$

C.3 Correlation Functions

The last step in solving a conformal field theory is finding the structure constants of the correlators, which we will do for the 2 and 3-point functions[22]:

$$\langle \mathbb{I}(z_1) \mathbb{I}(z_2) \rangle = 1 \quad (\text{C.36})$$

$$\langle \sigma(z_1) \sigma(z_2) \rangle = |z_{12}|^{-\frac{1}{4}} \quad (\text{C.37})$$

$$\langle \epsilon(z_1) \epsilon(z_2) \rangle = |z_{12}|^{-2} \quad (\text{C.38})$$

$$\langle \mathbb{I}(z_1) \sigma(z_2) \rangle = \langle \mathbb{I}(z_1) \epsilon(z_2) \rangle = \langle \epsilon(z_1) \sigma(z_2) \rangle = 0 \quad (\text{C.39})$$

Where $z_{ij} = z_i - z_j$. Now the 3-point functions:

$$\langle \mathbb{I}\mathbb{I}\sigma \rangle = \langle \mathbb{I}\mathbb{I}\epsilon \rangle = \langle \mathbb{I}\epsilon\sigma \rangle = \langle \sigma\sigma\epsilon \rangle = \langle \sigma\sigma\sigma \rangle = \langle \epsilon\epsilon\epsilon \rangle = 0, \quad (\text{C.40})$$

$$\langle \mathbb{I}(z_1)\mathbb{I}(z_2)\mathbb{I}(z_3) \rangle = 1, \quad (\text{C.41})$$

$$\langle \sigma(z_1)\sigma(z_2)\mathbb{I}(z_3) \rangle = |z_{12}|^{-\frac{1}{4}}, \quad (\text{C.42})$$

$$\langle \epsilon(z_1)\epsilon(z_2)\mathbb{I}(z_3) \rangle = |z_{12}|^{-2}, \quad (\text{C.43})$$

$$\langle \sigma(z_1)\sigma(z_2)\epsilon(z_3) \rangle = \frac{1}{2}|z_{12}|^{3/4}|z_{13}|^{-1}|z_{23}|^{-1}. \quad (\text{C.44})$$

Higher point functions can be exactly determined by the bootstrap method.